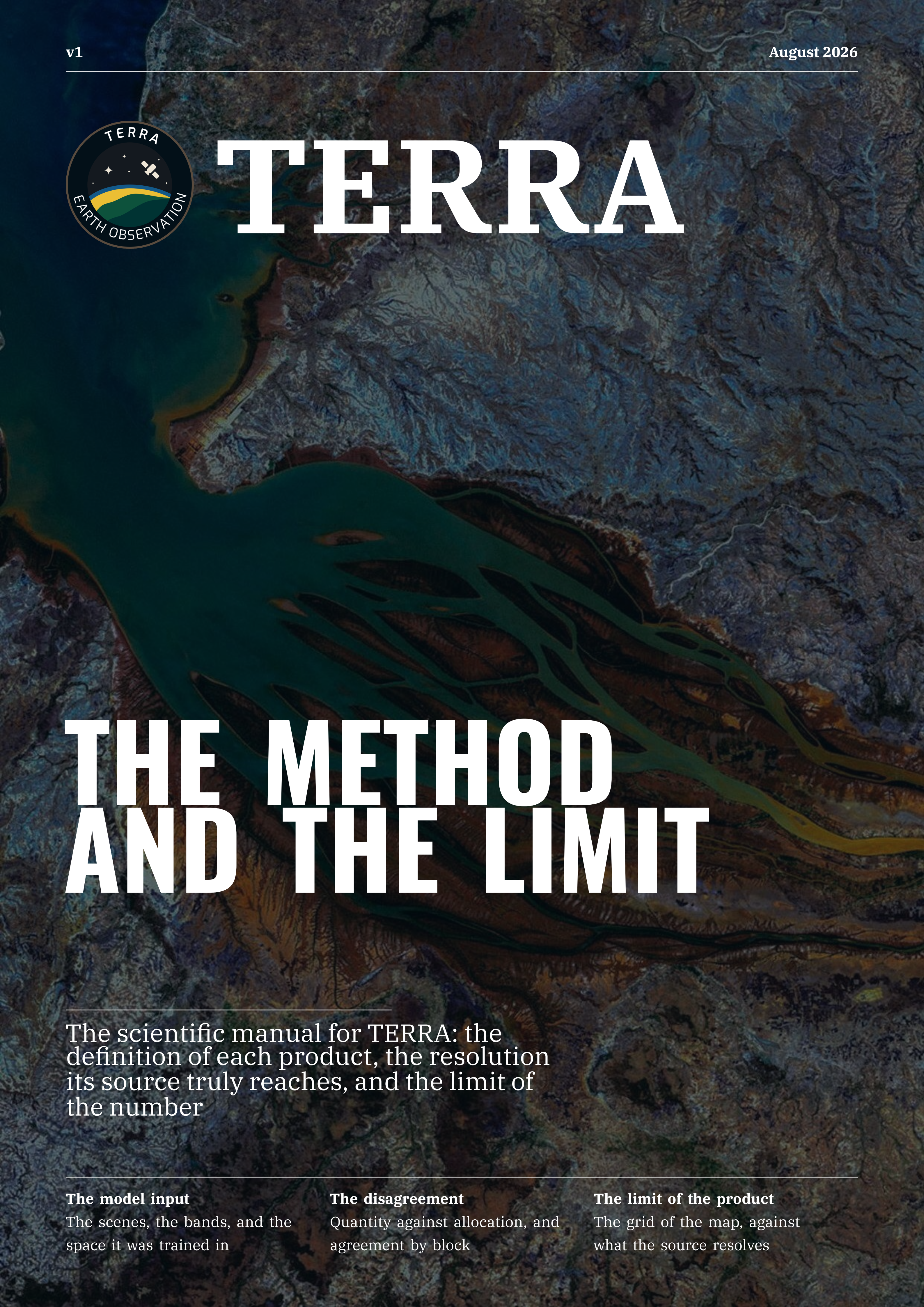

v1

August 2026

TERRA

THE METHOD AND THE LIMIT

The scientific manual for TERRA: the definition of each product, the resolution its source truly reaches, and the limit of the number

The model input
The scenes, the bands, and the space it was trained in

The disagreement
Quantity against allocation, and agreement by block

The limit of the product
The grid of the map, against what the source resolves

Contents

Preface

There are two questions people ask of a remote sensing tool, and they do not have the same answer. The first is *where do I click*. The second comes after the result appears on screen: *what exactly am I looking at?*

The TERRA *User guide* answers the first. It covers how an area is drawn, how the acquisition window is placed on the track, what each panel of the `Studio` shows, and what to do when the Python interpreter fails to import what the sidecar needs. It is a document of operation, and to operate the application it is enough.

This book answers the second. It exists for the person who, looking at a classified map, asks what the model is getting wrong and where; which spectrum it observed to decide that; whether a given number resolves the area that was drawn or the one-degree cell the area fell into. The screen shows the result. It does not show the definition of the metric, the true resolution of the source, or the point past which the number stops meaning what it appears to mean.

None of that fits in a tooltip. And none of it is optional for someone who will use the result rather than merely look at it.

Version described. TERRA 0.4.0, `9c8cad0`, of 23 August 2026. Where the text describes a behaviour of the code, it is the behaviour of that version. Where it describes a defect that is still open, it says so, and says what changes when the fix lands.

Who it addresses. Anyone who uses TERRA for an analysis and wants to understand what the application did with the data. It assumes familiarity with remote sensing and with reading a method; it does not assume knowledge of the application code. Where a formulation is cited, it comes with the reference and without the derivation.

Why it exists. For transparency. TERRA returns a map, a table and a figure, and almost every decision that produced those three is off screen: which scenes entered, in which space the model read the bands, what the agreement compares, where the source resolves and where it does not. A result whose construction cannot be inspected is a result that can only be accepted or refused.

What it does not cover. Installation and operation of the interface, which are in the *User guide*. No screen procedure is repeated here.

It also does not cover the full derivation of the methods, which stays in the study reports of the *TERRA-Research* line. Each chapter names the study rather than rewriting it. A manual that reproduces a derivation ages with it, and starts disagreeing with its source at the first revision.

How it is organised. Seven parts, one per product family, after a first part on what is common to all of them. Within each part, one chapter per question that is answered separately.

The appendices are for consultation: the catalogue of sidecar actions and the provenance table. They sit at the end because the reader returns to them with a specific question and does not read them in order. The short tables that support the argument of the paragraph beside them stayed where they were.

How each chapter is organised

All of them follow the same order, so that a reader who has read one knows where to look in the others:

The question. what the chapter decides, in one sentence.

Table 1: The division between the two manuals, question by question. The left column is the *User guide*; the right, this document.

User guide	This book
where the control is	what it computes
which value to type in the field	why the default is that value
how to read the legend	which unit the number is in, and over which sample
what the error message wants	what the true resolution of the source is
how to install and update	how far the number goes, and the reference that fixes it

The definition. the mathematics, where there is any, with the reference that fixed it. A formula and not a description: "*normalised water index*" does not say which band goes in the numerator, and two implementations that describe themselves with the same phrase can return different numbers.

What TERRA does with it. the parameters the application exposes, the default values, and what appears on screen as the result.

What cannot be claimed. the limit, with a number where there is a number.

The provenance of each statement does not stay in the chapter: the 119 entries are together in appendix B, with the file that implements it, the study that measured it or the reference that fixed it, and a column saying whether that source is public code, a study of the line, or literature. Sixteen lists split across sixteen chapters are one lookup, and whoever checks where a number came from looks for an entry, not for a chapter.

A manual that only says what the product does transfers to the reader the task of finding out where it stops. That discovery tends to arrive late, and through a result that did not close.

Conventions

English labels. the TERRA interface is in English, and labels are quoted as they appear on screen. They are never paraphrased either: `Build environment` is the name of a button, not a description of an action.

Decimal point. 3.5 cm in the body, `0.035` when the value is a literal from a file. The Portuguese edition uses a decimal comma in the body; this one does not.

Provenance with file and line. a reference such as `sidecar/flood.py:506` is from the *geosense-infer* repository, at the version declared above. Line numbers move; the function name, which the text always gives alongside, does not.

Studies by nickname. `E-hand-flood-baseline` is a study of the *TERRA-Research* line, and that is how the text refers to it. The nickname is stable; the internal organisation of the line is not, and it has been regrouped once already.

The line is private. *TERRA-Research* holds the studies that support what TERRA delivers, and it is not publicly distributed: the code, the intermediate data and the reports are under restricted access while the line works on them. A study named in this book can be requested, and section 1.1 says from whom. What is public is the application, under GPL v3, with the sidecar this book cites file by file.

List of abbreviations

Only what the text uses more than once. An abbreviation that appears once is defined at the point where it appears; this list serves whoever reads out of order.

BOA *Bottom of atmosphere*, the surface reflectance after atmospheric correction

CSI *Critical success index*, the name the flood literature gives to the Jaccard index between two binary masks

DEM *Digital elevation model*

DHI *Diffuse horizontal irradiance*, the diffuse irradiance on the horizontal plane

DN *Digital number*, the integer stored in the product, before any conversion to reflectance. It is how the text and the code write it, including in equation (3.1)

DNI *Direct normal irradiance*, the beam irradiance measured on a plane normal to the solar beam

GHI *Global horizontal irradiance*, the global irradiance on the horizontal plane, summing direct beam and diffuse sky

HAND *Height above the nearest drainage*, the height of a cell above the drainage it flows to

IoU *Intersection over union*, the Jaccard index between two masks. Numerically the same as the CSI

L2A The Sentinel-2 processing level that delivers surface reflectance. L1C, the level before it, delivers top-of-atmosphere reflectance

MERRA-2 *Modern-Era Retrospective analysis for Research and Applications, version 2*, the reanalysis the POWER meteorology comes from

NDVI *Normalized difference vegetation index*

POWER *Prediction of Worldwide Energy Resources*, the NASA service of meteorological time series

STAC *SpatioTemporal Asset Catalog*, the catalogue standard through which the application discovers scenes

SYN1DEG The CERES flux product, on a one-degree cell, which the POWER radiation comes from

Part I

The data and the protocol

CHAPTER 1

What TERRA delivers, and where it comes from

TERRA delivers five product families over the same drawn area: land cover, surface water, flood envelope, energy and canopy. They come from the same screen, in the same session, with the same appearance of a result.

They are not validated in the same way, and the difference appears nowhere in the interface. It determines, product by product, how far the number goes.

1.1 The line, and what reaches the product from it

TERRA is not where the methods are developed. It is the end of a research line, *TERRA-Research*, which runs the full cycle: literature review, a protocol registered before execution, simulation against acquired data, and a report.

$$\text{problem} \longrightarrow \text{research} \longrightarrow \text{result} \longrightarrow \text{TERRA}$$

The chain starts at a **problem**, not at an idea for a feature. The **research** attacks it under protocol, and what it produces is a **result**, which may be positive or negative. Only the last link is optional: a result is shipped once it has stabilised and once it puts something useful in the hands of whoever uses the application.

The line therefore has more results than the application has features. Three reasons, all of them legitimate:

The result is negative. a study that measures that a path does not work stops the application promising what it does not deliver. It is the most direct service the line renders to the product, and by definition it does not become a screen.

The result is good and has not been shipped yet. shipping costs interface, testing and packaging. The queue is decided by use, not by maturity.

The result is outside the scope of TERRA. the line answers questions the application does not set out to answer.

> *TERRA-Research* is private. Code, intermediate data and reports stay under restricted access while the line works on them.
> Every study named in this book can be requested, and the request needs no formal justification: write to `joao_leonardi.melo@somosicev.com` saying which study and what for. The application is public, under GPL v3, and it is the source of the file references in this book.

1.2 What validates each product

The question is not whether a dedicated study for the product exists. It is what supports the number. A published physical chain implemented faithfully supports as much as a study of one's own, and sometimes more. What changes between the rows below is the nature of the validation, and that is what fixes how far the number goes.

Two rows need careful reading, because they are the ones most open to being read past what they support.

Wind energy. the chain is published and the implementation is faithful, but nothing in it has been compared against an anemometer. Every turbine figure is gross of wake, availability,

Table 1.1: What validates the number of each product, and how far that validation carries it.

Product	What validates the number	How far it goes
Land cover	published protocol (Melo et al., 2026), and the agreement with MapBiomas measured on every run	a result, read with the reservations on fixed legend and on reference
Agreement and error	current definitions of thematic accuracy, computed over the run itself	a reproducible number, with the definition in this book
Surface water	three published indices and an Otsu threshold per date, with no trained model	the index, by the original source that defines it
Flood envelope	the disagreement between four elevation products, measured by the line in `E-hand-flood-baseline`	the envelope; the extent of a single product, no
Solar energy	a physical chain in `pvlib`, checked against the Global Solar Atlas, with the performance coefficient declared	quantified screening
Wind energy	a published physical chain, from Weibull to the reference turbine curve, with no comparison against wind measurement	screening, and only with the diagnostics alongside
Canopy	Beer-Lambert inversion, with the bias of the analytical path measured in `E-architecture-vs-slab`	comparison between scenarios, not an absolute value

electrical loss, icing, soiling and dispatch curtailment. And it rests on extrapolation above the highest level the reanalysis carries. The module carries that qualification in the result itself, and the application displays it beside the numbers. A TERRA wind result answers whether a site is worth measuring. It does not answer how much that site produces.

Flood envelope. the product is the envelope, not the extent. The study behind it measured that two $30\,\mathrm{m}$ elevation models disagree over about $21\,\%$ of the cells at the tightest threshold. Taking the extent of a single product is taking a shape that changes with the choice of product, and the screen does not say which part of it is terrain and which part is choice.

1.3 What the line has, and TERRA does not yet

People who use the application end up asking what else the line measures, and what has not reached the screen. The list below is the state in August 2026, and it changes between editions. When an item leaves it because it was shipped, the chapter for that product starts covering it. When it leaves because it was discarded, this section starts saying why.

The soil front is the most advanced of the open ones, and it is worth saying under which reservation. Field collection is closed, with 11,394 samples over 47 sites and 477 plots, between 2017 and 2024. Coverage is not uniform. The fatty acid groups and pH are present in $100\,\%$ of the samples; organic carbon and nitrogen, in $19\,\%$. That weakens exactly the set of variables the main hypothesis is tested against, and the conclusion will have to be read with that limit.

Table 1.2: Fronts of the *TERRA-Research* line with no corresponding feature in the application, with the state of each in August 2026.

Front	What it measures	State
Reservoir stage and volume without a gauge	water surface height by orbital altimetry, and an area-elevation curve paired with Sentinel-2 extents	closed, with a report; four chained studies
Phenological lag between regions	how many days one crop region is behind another, by two independent methods that agree	closed, with a result that reaches the user directly
Soil and microbial community	how much of the published relation between spectrum and soil microbiology survives controls for cover, geography and spatial autocorrelation	in progress, with a protocol registered before execution
Domain shift correction	six adaptation methods over a real pair of regions	closed, negative; the diagnosis entered TERRA, the correction did not
Roughness from an extent mask	whether the observed extent of a flood identifies the Manning coefficient	closed, negative

1.4 What TERRA is not

TERRA was built for the detailed study of particular areas, from the scale of a holding to that of a landscape, under a fixed protocol. It is not a general-purpose GIS. It does not cover the ground that Earth Engine and QGIS already cover.

Three consequences that arrive early for anyone coming from those tools:

The output legend is fixed. the crop models emit five MapBiomas classes and nothing else. An area in another biome can return a confident and semantically wrong result, and that is why the domain shift diagnosis exists in the product.

There is no map algebra and no layer editing. what the application computes is in the 17 sidecar actions, catalogued in appendix A. Extension is done by exporting the tables.

Nothing is sent to a server. the application runs locally, with no account and no service. Images are read on demand from the catalogue, only the window of the polygon and only the bands needed.

CHAPTER 2

The common protocol: area, catalogue and window

An area drawn in TERRA can be classified, have its water mapped, receive a flood envelope, a solar resource and a simulated canopy. Five products over one area, which it is natural to read as five readings of the same data.

They are not. Two of them read Sentinel-2, one reads four elevation models, and one reads a meteorological reanalysis on a one-degree cell. The fifth reads nothing new: it consumes what another has already produced. What follows is the part common to all five, and it is against this that each later chapter says where its own product departs.

2.1 The area of interest

The area is a polygon in `EPSG:4326`, drawn on the map or imported. It does three things, and in all three it is the polygon, not its bounding box, that governs:

It defines the catalogue query. the search is by intersection with the polygon.

It clips the read. bands are read only in the window of the polygon, not over the whole scene.

It bounds the statistics. every count, area, fraction and agreement is taken over the cells inside the polygon.

The third looks obvious and is not. In the flood envelope, the terrain chain needs the contributing area upstream, and therefore runs on a window *larger* than the area, 2 to 5 km beyond it. The report is still taken only inside the polygon.

Summarising the enlarged window instead of the polygon inflated every figure. On one observed run, a window of 277×291 cells of $30.8\,\mathrm{m}$ gave class areas summing to $76.2\,\mathrm{km}^2$, against an area of interest of about $20\,\mathrm{km}^2$. A factor of almost four.

2.2 The catalogue and the shape of the read

The optical source is the *Microsoft Planetary Computer*, collection `sentinel-2-l2a`, queried through **STAC**, the *SpatioTemporal Asset Catalog*. Bands are read on demand as *Cloud-Optimized GeoTIFF*: the window of the polygon and the bands needed, and nothing else. There is no download of a full SAFE product on any path of the application.

Table 2.1: The bands a classification run collects. A scene is not discarded when an extra band is missing; it only stops serving the paths that require it.

Group	Bands	For what
Required	B02, B03, B04, B08	the spectral Random Forest; without them the scene does not enter
Extra	B8A, B11, B12	the six channels of the Prithvi path and of the *Temporal Transformer*

The bands do not all arrive at the same native resolution. B02 to B08 are at $10\,\mathrm{m}$, and B8A, B11 and B12 at $20\,\mathrm{m}$. They are reprojected onto the reference grid of the run by bilinear resampling.

> Resampling B11 from 20 m onto a 10 m grid does not create 10 m information. A map drawn on that grid has a 10 m cell and 20 m content in the shortwave infrared bands. A feature smaller than 20 m appearing there comes from the interpolator. This holds for any product that uses the extra bands.

2.3 The acquisition window and the cadence

The window is the pair of dates the user drags on the track at the foot of the map. Two parameters go with it:

`max_cloud`. the maximum cloud cover accepted per scene, from the `eo:cloud_cover` field of the catalogue. The interface offers 40 % as the default; the sidecar, called directly, accepts up to 100 %.

`monthly_best`. on by default. It keeps, from each calendar month, only the scene with the least cloud cover.

The second is not a performance convenience. It brings the cadence of about one scene per month close to the set of 22 dates the models were fitted on. With it, the temporal statistics that feed the classifier stay comparable to those it saw in training.

Switching it off delivers every scene below the cloud limit. That is useful for a descriptive water series, and it is a mismatch for the classifier.

2.4 The 22 dates, and what happens outside them

The raw curve block of the spectral Random Forest has a fixed length. There are 22 positions, derived from the artefact itself: `n_dates_model` is the number of feature names minus 58, which is where the NDVI block starts. A window that produces a different number of dates is fitted to that length, and the fitting is not symmetric.

Table 2.2: What the application does when the window does not return exactly 22 dates. In both cases the model receives a vector of length 22 and has no way to tell a filled position from a measured one.

Dates obtained	What the code does	What the model receives
more than 22	keeps the **last** 22	the raw curve from the end of the window; the start is discarded
fewer than 22	pads with **zeros**	positions of value zero, indistinguishable from a measured NDVI of zero

> Widening the window does not give the classifier more raw curve. It gives band statistics and index descriptors computed over more dates, which use all of them, and an NDVI block that starts discarding the beginning of the period.
> Two runs over the same area with different windows do not differ only in the number of observations. They differ in *which* part of the series reaches the model raw. A comparison between them has to say so, or it attributes to the area what belongs to the window.

The path with fewer than 22 dates is the delicate one. A window short enough to yield fewer than 22 usable scenes produces a result the application does not refuse, and which the model reads as though the crop had NDVI zero over the missing stretch. There is no measurement, in this repository, of what that costs in accuracy. There is the observation that it happens, and that is enough to make checking the effective number of dates a step before reading the map. The application reports that number in the run bar and in the export package.

2.5 The neural paths read one scene, not the series

The *Temporal Transformer* and Prithvi do not consume the window the way the Random Forest does. Both take the scene from the middle of the period, literally the central element of the ordered product list, and Prithvi discards the rest.

The consequence has to appear in any method written with those paths: **widening the period does not give Prithvi more data, it changes which acquisition it reads**. A window from January to December and another from March to October can return different maps because they have different centres, with nothing having changed on the ground.

2.6 Which product reads what

Table 2.3: The sources by product, with the resolution each one resolves at. It is the resolution of the source, not the grid the result is drawn on.

Product	Source	Resolution of the source
Land cover	Sentinel-2 **L2A**, the level that delivers surface reflectance, through STAC	$10\,\mathrm{m}$; $20\,\mathrm{m}$ in the extra bands
Surface water	Sentinel-2 L2A, per date	the same
Flood envelope	four elevation products: `cop-dem-glo-30`, `nasadem`, `alos-dem`, `cop-dem-glo-90`	$30\,\mathrm{m}$ in the first three, $90\,\mathrm{m}$ in the fourth
Solar resource at a point	NASA **POWER**, *Prediction of Worldwide Energy Resources*: radiation from SYN1DEG, meteorology from MERRA-2	$1°$ for radiation; $0.5°$ by $0.625°$ for the rest
Irradiation over terrain, site	`cop-dem-glo-30` plus the POWER point	$30\,\mathrm{m}$ in the terrain, $1°$ in the radiation
Energy model	none new: it consumes the solar chain	—
Canopy	none new: it consumes the index series already computed	—

Two readings of that table matter more than the others.

The first: the energy products at a point do not resolve the area. POWER radiation is on a one-degree cell, and every point inside a holding-scale area returns the same series. The number describes the cell the area falls in, not structure within it. The terrain and site maps are the exception, and resolve at $30\,\mathrm{m}$.

The second: the two paths that read elevation do not read it the same way.

The flood envelope merges every tile intersecting the enlarged window. A drainage network cut at the edge of the area produces high HAND and a small extent: a plausible map, wrong in a direction the screen does not reveal.
The solar chain does not do this. `solar.fetch_dem` reads `items[0]`, the first tile returned, with no merge. Copernicus tiles are one degree, so an area crossing a border receives an elevation model that covers part of it. For slope and aspect this degrades in patches. It is a limitation of the irradiation-over-terrain and photovoltaic site maps, not of the flood envelope.

2.7 What cannot be claimed

That two products over the same area are over the same data. they are over the same geometry. Table 2.3 is the list of sources that differ.

That the grid of the map is the resolution of the data. a water map drawn on a $10\,\mathrm{m}$ cell using MNDWI carries $20\,\mathrm{m}$ content in band B11. An irradiation-over-terrain map on a $30\,\mathrm{m}$ cell carries one-degree-cell radiation modulated by $30\,\mathrm{m}$ geometry.

That widening the window always improves the result. for the Random Forest, it changes which stretch of the series arrives raw. For Prithvi, it changes which acquisition is read. For water, it increases the number of dates, and with it the denominator of the persistence fractions.

That the number of scenes is fixed by the protocol. it is fixed by the window, the cloud limit, catalogue availability and the monthly filter. Any method written with a TERRA result declares all four.

CHAPTER 3

From the stored number to the number the model sees

Sentinel-2 does not distribute reflectance. It distributes an integer digital number, and reflectance comes out of a conversion. Inside TERRA that conversion has two versions, one physical and one historical, and both are in use at the same time, on purpose.

Anyone about to read a spectral value from TERRA needs to know which of the two produced the number on screen. The difference between them is 0.1 in every band, and nothing in the interface indicates it.

3.1 The physical conversion

Processing baseline 04.00 applies to products generated from 25 January 2022. In it, Sentinel-2 shifts the dynamic range by a constant, so that reflectance near zero stops being clipped over dark surfaces (ESA, 2021). The shift is stored in the product metadata. Therefore:

$$\rho = \frac{\mathrm{DN} + \mathrm{BOA\_ADD\_OFFSET}}{10000}, \qquad \mathrm{BOA\_ADD\_OFFSET} = -1000 \tag{3.1}$$

Reading the digital number without the offset term overestimates *every* band by 0.1. Some catalogues harmonise this on delivery. The Planetary Computer, which is what TERRA reads by default, does not: its items expose neither scale nor offset in `raster:bands`, and the correction is left to the application.

The offset is derived per product, not fixed as a constant. A scene from before the change does not carry it, and subtracting an offset from that scene would introduce exactly the error the function exists to remove. The decision uses `s2:processing_baseline` where the catalogue reports it, and falls back to the acquisition date where it does not.

Everything the application *reports* as a physical quantity passes through here: the vegetation indices, the phenology, the water masks, the band composites and the series that feeds the canopy.

3.2 The space the models learned in

The artefacts in `model/` were not fitted in that space. They were fitted on inputs built as $\mathrm{DN}/10000$, without the offset term. The set is 22 Sentinel-2 products acquired between 4 May 2024 and 4 January 2026, over tiles T21JZN and T22JBT.

All of those products are later than baseline 04.00. That is: the training image carried the offset and the training pipeline did not subtract it. The heads learned a space in which every band is 0.1 high, and they are coherent inside it.

> Nothing was mismatched. **Correcting inference alone is what would create the mismatch**, and that is measured. Over an area in Cascavel, converting the model inputs with the offset moves $56.8\,\%$ of the pixels, and turns $70.8\,\%$ soybean into $62.0\,\%$ forest formation over cropland. That is not a gain in accuracy. It is a model answering a question it was never asked.

The seam is therefore deliberate, and explicit in the code. There are two functions: `to_reflectance` for what is reported, and `as_trained` for the three model inputs. The seam closes when the heads are refitted on corrected input, and at that moment `as_trained` is deleted, not changed.

Table 3.1: The two conversions, with the formula of each and who consumes it.

Function	Formula	Consumers
`to_reflectance`	$(\text{DN} + \text{offset})/10000$	vegetation indices, phenology, water masks, composites and the canopy series: everything the screen calls reflectance
`as_trained`	$\text{DN}/10000$	the three model inputs: spectral Random Forest, *Temporal Transformer*, Prithvi

3.3 What this changes in the reading

A reflectance value read in the spectral panel is not the classifier input. for a post-baseline scene, the model input is 0.1 above the displayed value, in every band. Comparing a number from the panel against a threshold taken from the model literature is legitimate; comparing it against what the head received is not.

Normalised indices are not saved by the normalisation. an NDVI computed with the offset and one computed without do not differ by a factor. They differ by a term that does not cancel: the offset enters the numerator and the denominator equally, and not the result. The NDVI the application reports uses the physical conversion; the raw NDVI block the Random Forest consumes does not.

Separability and domain shift inherit the training space. the fingerprints written at classification time live in the space of `as_trained`. That is the right choice for comparing two runs with each other, and it is the reason those numbers must not be read as reflectance.

3.4 The open defect: the scene from before 2022

`as_trained` receives only the digital number. It does not receive the product, and therefore does not distinguish a pre-baseline scene from a post-baseline one. For a scene from 2022 onwards that is exactly what is wanted. For an earlier scene, it is silently wrong by -0.10.

Table 3.2: Same point, same read path, two years. The difference is the offset, and it is of the order of 0.1 in every band. A 2021 scene handed to a head trained in the 2024 space enters displaced by that amount.

Band	2021	2024	Δ
B02	$0.019{,}3$	$0.122{,}6$	$0.103{,}3$
B03	$0.035{,}5$	$0.139{,}3$	$0.103{,}8$
B04	$0.020{,}2$	$0.122{,}8$	$0.102{,}6$
B8A	$0.300{,}8$	$0.406{,}7$	$0.105{,}9$
B11	$0.148{,}8$	$0.250{,}1$	$0.101{,}3$
B12	$0.062{,}8$	$0.161{,}2$	$0.098{,}4$

The artefact itself confirms that the training space is the shifted one and not the physical one. The mean of the valid pixels of the three training stacks is $0.148{,}0$, $0.172{,}5$, $0.184{,}8$, $0.400{,}7$, $0.332{,}2$ and $0.253{,}6$, and the mean of the scaler stored in the checkpoint is $0.150{,}7$, $0.175{,}5$, $0.188{,}4$, $0.404{,}2$, $0.338{,}2$ and $0.259{,}5$.

The size of the problem comes from that same scaler. With a standard deviation of $0.030{,}9$ in the first position, feeding the *Temporal Transformer* a 2021 scene without lifting it into the training space puts the input about **three standard deviations** from what it learned in the blue.

> In TERRA 0.4.0 this is not fixed: `as_trained` still takes one argument. The proposed fix is additive. It passes the product and cancels the term when the product is already post-baseline, so that post-baseline behaviour stays identical to today. It is described in a correction note of the *TERRA-Research* line, with the test that covers both sides of the baseline.

The rule until then. a classification whose window falls before 25 January 2022 has to be read knowing this. The product gives no warning, the map comes out plausible, and the error is three standard deviations in the most sensitive band of the most recent model. In a window that crosses the date, some scenes are in the right space and some are not, and the inconsistency is internal to the series.

3.5 What cannot be claimed

That the seam is an error. the measurement says the opposite, and it is in section 3.2. What is an error is the pre-baseline case, and only that.

That the two numbers are interchangeable through one calculation. they are, for a single scene and a single band: 0.1. They are not, for any derived quantity, whether a normalised index, a phenological descriptor or a distance between distributions. The difference does not cross the derivation as a constant.

That the offset was harmonised by the catalogue. it was not, in the Planetary Computer. A pipeline reading the same product from another catalogue may receive values already harmonised. In that case the conversion of equation (3.1) would apply the term twice.

Part II

Land cover classification

CHAPTER 4

The three model paths, and the fixed legend

The screen offers three models and two modes, and the choice between them changes the map. It also changes what the result means, because the three do not read the same slice of the data that chapter 2 described. One consumes the whole series in summarised form, one consumes the series as a sequence, and one consumes a single acquisition.

What the three have in common is the output vocabulary, and it is that vocabulary which imposes the hardest limit in the application.

4.1 The five classes, and why there are five

The three paths emit the same five MapBiomas codes and nothing else. The vector is stored in the artefact itself, in `model/label_encoder.joblib`:

Table 4.1: The five classes any of the three paths can emit, with the corresponding MapBiomas code. A pixel receives one of these, always.

Code	MapBiomas class	How to read it
3	Forest formation	tree vegetation, native or not
21	Mosaic of uses	agriculture and pasture mixed within the cell
25	Non-vegetated area	bare soil, urban area, rock outcrop
39	Soybean	the only crop with a class of its own
41	Other temporary crops	everything else that is a short-cycle crop

> Class 41 is a remainder, not a crop. Maize, wheat, beans, oats and any other temporary crop all fall into it, without distinction. A high overall accuracy against MapBiomas does not imply that the model can say which crop is which. It implies that the model gets the boundary right between soybean, the rest of the crops, mosaic, forest and bare soil. That boundary is what the agreement measures, not the identity of the crop inside class 41.

What happens outside the vocabulary. nothing visible. An area of pure pasture, savanna, mangrove or sugarcane receives one of the five codes, with confidence, because the classifier has no way of answering "*none of the above*". It is the failure mode the screen does not report, and it is why the domain shift diagnosis exists. That diagnosis does not correct the wrong class. It measures whether the drawn area resembles the one the model was fitted on.

4.2 The `spectral` path: 80 features per pixel

It is the application default. Each pixel becomes a vector of 80 numbers, and figure 4.1 shows how those 80 divide.

The 14 metrics. are, in the order in which they appear in the vector: mean, standard deviation, maximum, minimum, range, median, index of the peak, index of the minimum, wet season mean, dry season mean, seasonal difference, mean variation between dates, largest increase between consecutive dates, and largest drop between consecutive dates.

The three indices they are computed over are defined as follows, with ρ the reflectance of the band indicated:

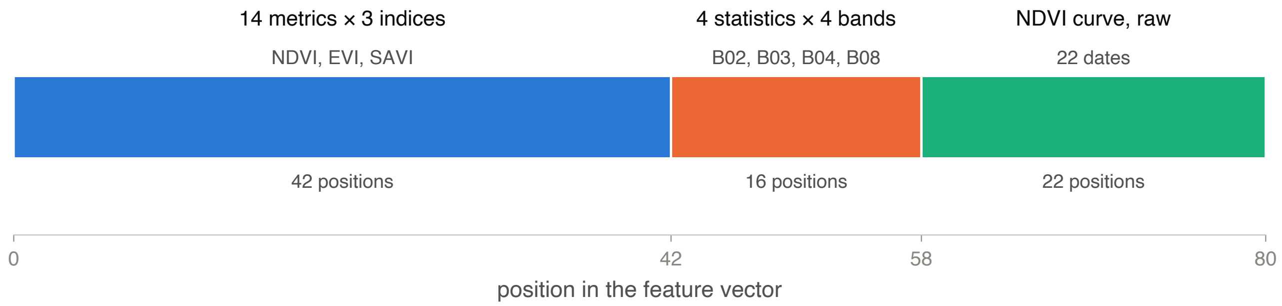

Figure 4.1: The feature vector of the `spectral` path, read from `model/feature_names.joblib`. The bar has 80 positions, from 0 to 79, and is divided into three contiguous blocks. **Blue, positions 0 to 41:** the 14 temporal metrics, repeated for each of the three vegetation indices, in the order NDVI, EVI, SAVI. **Orange, positions 42 to 57:** four statistics, mean, standard deviation, maximum and minimum, for each of the four $10\,\mathrm{m}$ bands, in the order B02, B03, B04, B08. **Green, positions 58 to 79:** the NDVI curve with no summary at all, one position per date, for the 22 dates of the protocol. The cuts on the horizontal axis mark the boundaries between blocks, at 42 and at 58. The reading of the figure is the proportion: more than half the vector, 64 of the 80 positions, describes NDVI, first compressed into metrics and then repeated raw.

$$\mathrm{NDVI} = \frac{\rho_{\mathrm{B08}} - \rho_{\mathrm{B04}}}{\rho_{\mathrm{B08}} + \rho_{\mathrm{B04}}} \tag{4.1}$$

Equation (4.1) is the normalised difference between the near infrared, B08, and the red, B04. A living leaf reflects strongly in the near infrared and absorbs in the red, so the numerator grows with the amount of leaf and the denominator normalises by the total brightness of the scene. The result lies between -1 and 1: water and shadow give negative values, bare soil sits near zero, and a closed canopy passes 0.8.

$$\mathrm{EVI} = 2.5 \cdot \frac{\rho_{\mathrm{B08}} - \rho_{\mathrm{B04}}}{\rho_{\mathrm{B08}} + 6\,\rho_{\mathrm{B04}} - 7.5\,\rho_{\mathrm{B02}} + 1} \tag{4.2}$$

Equation (4.2) has the same numerator as NDVI and a different denominator, and that is where the point is. The term $6\,\rho_{\mathrm{B04}}$ and the $-7.5\,\rho_{\mathrm{B02}}$ use the blue to estimate atmospheric scattering and discount it. The $+1$ avoids division by zero over a dark surface, and the factor 2.5 rescales the result to a range comparable with NDVI. The practical effect is that EVI saturates later than NDVI over a dense canopy, where NDVI has already reached its ceiling and stops distinguishing.

$$\mathrm{SAVI} = \frac{\rho_{\mathrm{B08}} - \rho_{\mathrm{B04}}}{\rho_{\mathrm{B08}} + \rho_{\mathrm{B04}} + L} \cdot (1 + L), \qquad L = 0.5 \tag{4.3}$$

Equation (4.3) is NDVI with a soil term. The L in the denominator damps the contribution of substrate brightness, which is what makes NDVI vary over a still sparse crop even when the foliage has not changed. The factor $(1 + L)$ returns the result to the NDVI scale.

With $L = 0.5$, a value in current use for intermediate cover, SAVI and NDVI converge over a closed canopy. They diverge over partially exposed soil, which is the crop stage at which the distinction between crops matters most.

> These three indices enter the feature vector computed in the training space, not in the physical space. It is the seam of chapter 3: the NDVI the application *reports* uses `to_reflectance`, and the NDVI block the model *consumes* uses `as_trained`. The two are not the same number, and the difference is not constant along the derivation.

4.3 The other two paths

`Temporal Transformer`. consumes the series as a sequence and not as a summary: 22 frames, six bands per frame, in the order B02, B03, B04, B8A, B11 and B12. It is the only one of the three shipped paths that sees the shortwave infrared, B11 and B12, which chapter 2 describes as already downloaded but not used by the default classifier. The output is reduced by averaging over time before the decision.

`Prithvi-EO 2.0`. uses a frozen foundation model of 300 million parameters to turn one acquisition into a vector of 1024 dimensions, and decides over that vector with a Random Forest. It takes the scene from the middle of the period and discards the rest. Chapter 2 has already qualified that: widening the window does not give it more data, it changes which acquisition it reads. There are two modes, `pixel` and `patch`, which differ in the unit the embedding is taken over.

The `Temporal` mode. accumulates dates and reports what stays classified across the period, instead of a single map. It exists only in `spectral`: the other two produce the map and nothing to accumulate, and the interface refuses the combination saying so.

4.4 What the measurement by the line says about the choice

The line measured how different representations of the same pixel perform against real MapBiomas. Figure 4.2 carries the result, and it is not a single ordering.

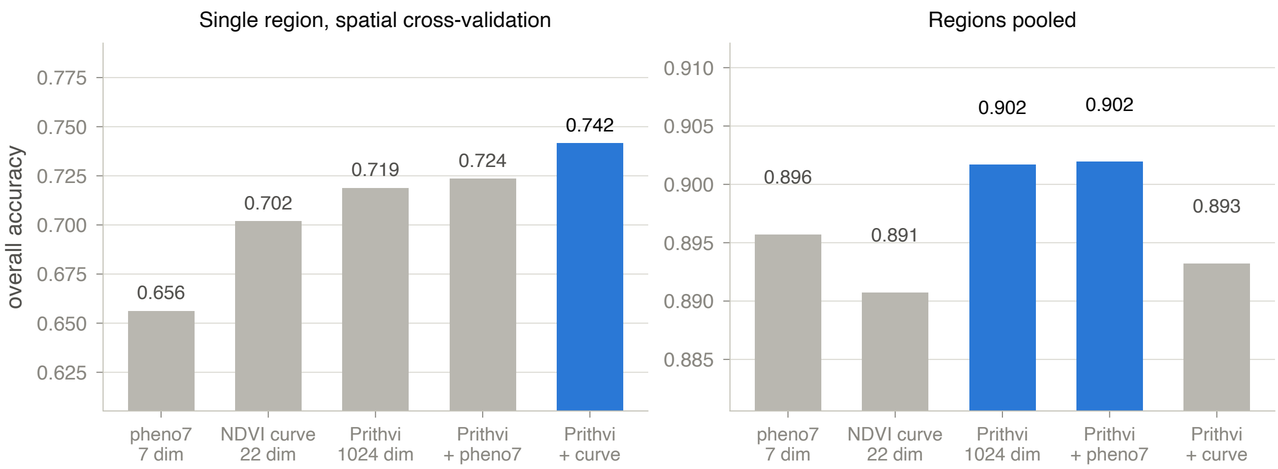

Figure 4.2: Overall accuracy of five representations of the same pixel, in two validation scopes. Each panel has the same five bars, in the same order from left to right: `pheno7`, seven phenological metrics; the raw NDVI curve, 22 dimensions; the Prithvi embedding, 1024 dimensions; and the two combinations of the embedding with each of the previous two. The value of each bar is printed above it, and the bar in blue is the one that leads the panel. **Left panel:** a single region with spatial block cross-validation, $n = 832$, which is the condition closest to real use of the application. **Right panel:** pooled regions, $n = 4{,}018$. The vertical scales are independent, and deliberately so: the right panel operates between 0.885 and 0.910, and the left between 0.625 and 0.775, so a shared axis would flatten the internal comparison in each. What the figure asks the reader to compare is the *order* within each panel, not the level between panels. On the right, two bars are highlighted because they tie at the third decimal place.

Three readings, and the third is the one that matters for whoever picks a model on screen.

The raw curve beats the summary. in the left panel, the 22 positions of the NDVI curve give 0.702 against 0.656 for the seven phenological metrics. Compressing the series into descriptors costs accuracy, and the `spectral` path carries both forms for exactly that reason.

Not every extra dimension helps. the 1024-dimension embedding gives 0.719, and adding the seven metrics to it reaches 0.724. Adding the raw curve reaches 0.742. The gain comes from which information enters, not from how many dimensions.

The order inverts between the scopes. on the left, the combination with the raw curve leads at 0.742. On the right, it is *last*, at 0.893, below the embedding on its own. What leads on the right is the embedding, pure or added to the metrics. The two tie at 0.902, and separate by $0.000{,}2$ over $4{,}018$ samples. That is less than one sample.

> The inversion is not noise, it is the effect of the validation. Mixing the regions puts part of the neighbourhood of each test block inside the training set, and accuracy rises for every representation, from about 0.70 to about 0.89. A gain that appears only in that condition is a gain the spatial validation does not confirm. **The left panel is the one that resembles use of the application**: a model fitted in one place, applied to an area the user drew in another.

Between the two neural models, the same line measured the Temporal Transformer against a perceptron over the Prithvi embedding. Under spatial validation in a single region, the perceptron gives 0.737 and the Transformer 0.708. Having the raw series and the shortwave infrared was not enough for the Transformer to lead.

4.5 What cannot be concluded

That figure 4.2 orders the three buttons on screen. it does not. The ablation measures *representations*, and the `spectral` path is a fourth arrangement, of 80 dimensions, that is not in it. What can be said is that the two blocks it is made of, the metrics and the raw curve, measure below the Prithvi embedding in both columns where they appear.

That the best combination is available. it is not. The one that leads the spatial validation is the embedding added to the raw NDVI curve, $1{,}046$ dimensions. It is not one of the three shipped paths: the Prithvi artefacts in `model/` carry the $1{,}024$ dimensions of the embedding and nothing more.

That overall accuracy speaks for your area. it is measured over the validation set of the line, in western Paraná. For the area you drew, the number to look at is the agreement the run itself computes, not these. The chapters of the next part deal with that.

That changing the model changes the legend. it does not. The three emit the same five codes, and a different model can change which of the five falls on each pixel. None of them widens the vocabulary.

CHAPTER 5

What the map claims, and in which unit

The run has finished. There is a coloured map over the area, a list of classes with a percentage and a hectare figure beside it, and a confidence value. Each of those numbers was taken over a specific sample, and none of them says what the immediate reading suggests.

This chapter is about the distance between those two things.

5.1 What the run returns

Table 5.1: The numeric fields a classification returns, with the sample each one is taken over. The right column is what the chapter details below.

Field	What it is	Over which sample
`class_stats`	per class: pixel count, percentage and hectares	the *classified* pixels, not the drawn area
`mean_confidence`	mean of the per-pixel confidence	the same, and on a scale that does not start at zero
`confidence_floor`	the smallest value confidence can take	a constant, $1/K$, with K the number of classes
`pixel_size_m`	the side of the cell, read from the grid	the run
`n_dates`, `date_range`	how many scenes entered and the period they cover	the run
`temporal`	one entry per date, in `Temporal` mode	the stack accumulated up to that date

5.2 The class fractions, and the denominator

Each pixel receives one of the five classes, or the value -1 when it could not be classified. The percentage of each class is computed as:

$$\mathrm{pct}_c = 100 \cdot \frac{n_c}{\sum_j n_j} \tag{5.1}$$

In equation (5.1), n_c is the pixel count of class c, and the denominator sums the counts of *all* classes. The point is in what the denominator does not include: pixels marked -1 stay outside both sides of the fraction.

> The percentages sum to $100\,\%$ of what was classified, not of the area you drew. Cloud, shadow or missing data remove part of the polygon, and that part appears nowhere in the list. It does not become a "*no data*" class: it leaves the calculation. A reading of $70\,\%$ soybean over a polygon half of which was discarded describes $70\,\%$ of the half that remained.

The number that closes that calculation is the `n_dates` of chapter 2, together with the pixel count. The sum of `pixels` over all classes, multiplied by the cell area, gives the area actually classified. Comparing that area against the area of the polygon is the check, and the application does not perform it.

5.3 The hectare is a pixel count

The area of each class comes from the count, with the cell worth $100\,\mathrm{m}^2$:

$$\text{area\_ha}_c = n_c \cdot \frac{100}{10^4} \tag{5.2}$$

Equation (5.2) converts pixels into hectares by a fixed factor. The 100 in the numerator is the area of one cell in square metres, from a grid $10\,\mathrm{m}$ on a side. The 10^4 in the denominator is the number of square metres in a hectare.

For the classification path the 100 is correct: the reference grid is built from band B04, at its native resolution of $10\,\mathrm{m}$.

Two readings follow from that, and both matter.

The area is not corrected for classification error. if the model confuses soybean with another temporary crop in some of the pixels, the soybean hectare figure carries that error whole. The conversion is arithmetic over the count, not an area estimate with uncertainty. Area estimators that correct through the confusion matrix exist, and the application does not apply them.

The factor is fixed, and the measured size travels beside it. the same payload that carries `area_ha` carries `pixel_size_m`, which is read from the transform of the grid instead of assumed. Today the two agree. If the reference grid ever stops being $10\,\mathrm{m}$, the hectare does not follow the measured size, and nothing in the output flags the divergence. When checking a published area, check the `pixel_size_m` of the run.

5.4 Confidence is not probability of being right

The confidence value of a pixel is the largest component of the probability vector the classifier returns:

$$\text{conf} = \max_{c \in \{1..K\}} p_c, \qquad \sum_{c=1}^{K} p_c = 1 \tag{5.3}$$

In equation (5.3), p_c is the probability assigned to class c and K is the number of classes, five in this product. In the Random Forest, p_c is the fraction of trees that voted for that class; in the *Temporal Transformer*, it is the corresponding component of the softmax of the output. In both cases confidence measures **internal agreement of the model**, not the rate of correct answers.

From that follows the property that most often misleads the reading. A vector of K components summing to 1 has a maximum no smaller than $1/K$, so confidence lives in $[1/K,\ 1]$ and never approaches zero. With five classes, the floor is 0.20. Figure 5.1 shows the effect on the scale.

The displacement between those two readings is what the code itself records. Over five classes, $38\,\%$ is about one fifth of the way from maximum uncertainty to certainty, not one third. That is why the payload carries `confidence_floor` beside the mean: without K, the consumer has no way of repositioning the number.

> High confidence does not protect against the fixed legend of chapter 4. Over an area of pasture, which is not in the vocabulary, almost all the trees can agree on "*mosaic of uses*" and produce a confidence of 0.9. What that value says is that the model did not hesitate. It cannot say that the model was right, because the right answer was not among the options.

5.5 The `Temporal` mode

In `Temporal` mode, the application does not return a map. It returns a sequence: for each date in the window, it reclassifies using the stack accumulated up to that point, from the first scenes to

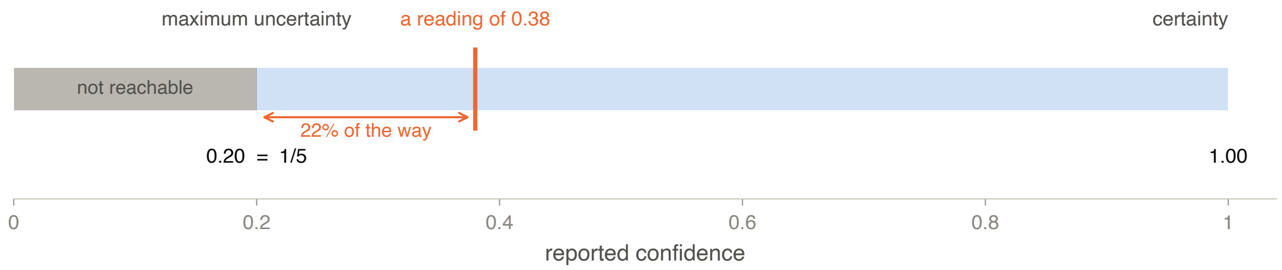

Figure 5.1: The scale the reported confidence lives on, for the five classes of this product. The horizontal axis runs from 0 to 1, which is the scale the value appears on in the interface. **The grey block on the left, from 0 to 0.20, is unreachable:** no pixel can receive a confidence below $1/K$, because the maximum of five probabilities summing to 1 does not fall below it. **The light blue block, from 0.20 to 1.00, is the range that exists.** Its left end is maximum uncertainty, with the five classes tied at 0.20 each, and its right end is certainty of the model. **The orange line marks a reading of 0.38**, and the arrow below it measures where that reading falls inside the range that exists: $22\,\%$ of the way between maximum uncertainty and certainty. Read on the 0 to 1 scale, the same reading would appear to be at $38\,\%$.

that one, and reports the result.

Each entry carries the date, how many scenes entered the stack, the dominant class, and soybean retention where there is a MapBiomas raster to compare against. The reading is one of stability. A class that only appears once the stack is nearly complete depends on the whole series; one that settles early is decided by the start of the season.

> Each step of `Temporal` mode is a complete run of the classifier over the stack up to that date, not a refinement of the previous step. The truncation to 22 dates of chapter 2 applies at every step: the first steps hand the model a vector padded with zeros in the positions that do not yet have a date.

5.6 What cannot be concluded

That the percentages describe the polygon. they describe the pixels that received a class. The fraction of the polygon left without data does not appear in the list.

That the hectare is an area estimate. it is a converted count. It carries no correction for classification error and no interval.

That high confidence indicates a correct answer. it indicates agreement among the trees, or mass concentrated in the softmax. Over an area outside the vocabulary, both coexist with a wrong answer.

That $38\,\%$ confidence is a little over a third of the scale. it is about one fifth of the usable range. The scale starts at 0.20, not at zero, and the `confidence_floor` of the run carries that floor.

That the dominant class of `Temporal` mode converges. nothing guarantees convergence. The sequence can oscillate between classes as dates enter, and the oscillation is the result, not a defect in the reading.

Part III

Error, agreement and transfer

CHAPTER 6

Agreement with MapBiomas: the matrix and its intervals

When the drawn area intersects Brazil, the run compares the classification with MapBiomas and reports an agreement. It is the number that answers the question *where is it getting things wrong*. It is also the one most open to being read past what it measures.

This chapter says what was compared with what, over how many observations, and what was left out of the count.

> MapBiomas is not ground truth. It is an annual map, produced by another classifier, and generally out of step with the series of dates the run read. What the application measures is **agreement** between two maps, and a disagreement can be an error on either side. The text of this book says agreement everywhere for that reason.

6.1 The sampling unit is the cell, not the pixel

MapBiomas is native at $30\,\mathrm{m}$ and is resampled onto the $10\,\mathrm{m}$ grid of the classification. About nine pixels then carry **one** label observation, repeated.

The application aggregates the comparison back to the native cell. Each cell contributes one observation: the reference label it carries, against the majority predicted class among its pixels. The mode protects the cell that falls on a resampling boundary, instead of assuming that the nine pixels agree.

The alternative would be to count pixels, and it is attractive because it multiplies the sample by nine with no work at all. Figure 6.1 shows what that does to the interval.

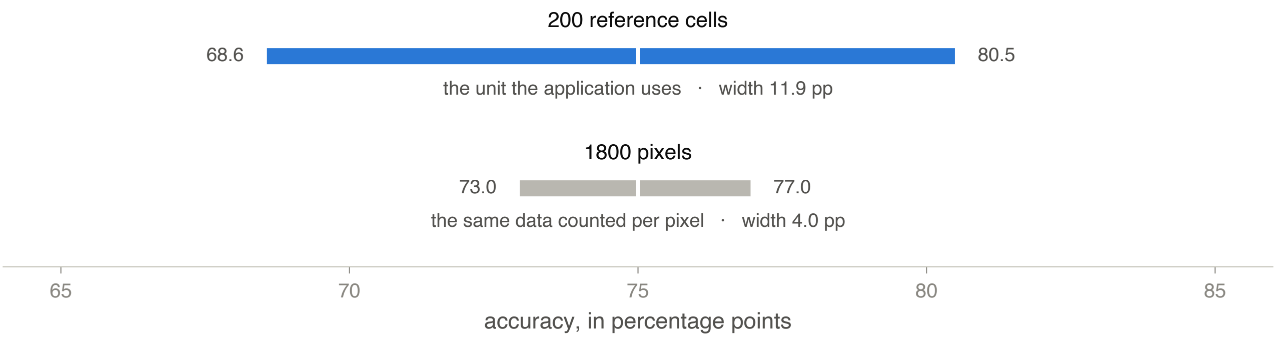

Figure 6.1: The $95\,\%$ confidence interval for the *same* accuracy of $75\,\%$, computed over two counts of the same data. The horizontal axis is accuracy in percentage points. **The blue bar, above,** is the interval over 200 reference cells, which is the unit the application uses: it runs from 68.6 to 80.5, a width of 11.9 points. **The grey bar, below,** is the interval over the $1{,}800$ pixels those same 200 cells occupy on the $10\,\mathrm{m}$ grid: it runs from 73.0 to 77.0, a width of 4.0 points. The white tick in the middle of each bar marks the point estimate, identical in both. The reading is the ratio between the widths, 2.98: counting pixels instead of cells narrows the interval by a factor of very nearly three, without a single new label observation having been made.

The factor of three is not a coincidence. The half-width of the interval falls with $1/\sqrt{n}$, and multiplying n by nine divides it by $\sqrt{9}$.

When the cell mapping is unavailable, the application returns *nothing* instead of returning the per-pixel count. The choice is recorded in the code: an accuracy with the wrong n is worse than no accuracy, because it still gets believed.

6.2 The matrix, and the three accuracies

The comparison builds a matrix M of five by five, with the predicted classes in the rows and the reference classes in the columns. M_{ij} is the number of cells the classifier called i and the reference calls j. The correct cells lie on the diagonal.

Three numbers come out of it, and they answer different questions.

$$\mathrm{OA} = \frac{\sum_i M_{ii}}{n}, \qquad \mathrm{PA}_c = \frac{M_{cc}}{\sum_i M_{ic}}, \qquad \mathrm{UA}_c = \frac{M_{cc}}{\sum_j M_{cj}} \tag{6.1}$$

In equation (6.1), n is the total number of cells compared.

OA, overall accuracy. the fraction of all cells in which the two maps say the same thing. It is the sum of the diagonal over the total. It does not distinguish a large class from a small one: in an area dominated by one class, getting only that class right already delivers a high overall accuracy.

PA_c, producer accuracy. looks at a *column*. Of the cells the reference calls c, the fraction the classifier found. The complement is omission: what existed and went unnoticed.

UA_c, user accuracy. looks at a *row*. Of the cells the classifier called c, the fraction the reference confirms. The complement is commission: what was called c and was not.

The last two do not move together, and the difference between them is diagnostic. A classifier that calls almost everything soybean lets almost nothing through, so the producer accuracy for soybean is high. The user accuracy is low, because most of what it called soybean is not. The reverse describes a timid classifier.

Table 6.1: The three accuracies, with the question each one answers and the error its complement measures.

Measure	Reads the matrix by	Answers	The complement is
Overall	the whole diagonal	in what fraction of cells the two maps agree	the total disagreement
Producer	column, per class	of the class that exists, how much was found	omission
User	row, per class	of what was called this, how much actually is	commission

6.3 The interval is Wilson, not Wald

Each of the three accuracies comes with a $95\,\%$ interval. The interval is the Wilson one:

$$\mathrm{CI} = \frac{\hat{p} + \dfrac{z^2}{2n}}{1 + \dfrac{z^2}{n}} \pm \frac{z}{1 + \dfrac{z^2}{n}} \sqrt{\frac{\hat{p}\,(1-\hat{p})}{n} + \frac{z^2}{4n^2}} \tag{6.2}$$

In equation (6.2), $\hat{p}$ is the observed proportion, n is the number of cells and $z = 1.96$ for $95\,\%$. The centre of the interval is not $\hat{p}$. It is $\hat{p}$ pulled towards 0.5 by a term that depends on n, and that displacement shrinks as the sample grows. The root is the standard deviation of the proportion, with a correction of the same kind.

That form is a choice, and the alternative is the Wald interval, $\hat{p} \pm z\sqrt{\hat{p}(1-\hat{p})/n}$, which is what most spreadsheets compute. Wald has two defects under the conditions of this product. With accuracy near $95\,\%$ and a small sample, it produces limits above 1 or below 0, which are not proportions. And it covers less than it promises: an interval said to be $95\,\%$ contains the true value in fewer than $95\,\%$ of cases. Wilson stays inside the unit interval and holds the nominal rate at small n (Brown et al., 2001).

6.4 What is left out of the count

Not every reference cell enters the matrix. The drawn area can contain water, urban area, forestry or any other MapBiomas class not among the five of chapter 4. Those cells are counted separately, in the field `n_outside_legend`, and do not enter as error.

The reason is that counting them as error would merge two different things: *classified wrongly* and *outside the vocabulary*. A classifier with no label for water did not err by failing to say water; it was not asked.

> `n_outside_legend` is the fraction of the area about which the evaluation says nothing. It appears neither in the overall accuracy nor in any of the per-class accuracies. An agreement of $88\,\%$ over an area where a third of the cells fell outside the legend is $88\,\%$ of the remaining two thirds. Check that field before reading any of the others.

6.5 Kappa is not reported

The application does not return the kappa coefficient, and the absence is deliberate. Its place is taken by two readings the payload already carries. One decomposes the disagreement into quantity and allocation; the other distributes it over space, in blocks. The next two chapters develop them.

6.6 What cannot be concluded

That the agreement is accuracy. it is agreement between two maps, and one of them also errs. Without field verification, a disagreement does not say which side the error is on.

That overall accuracy describes the classes. it describes the class mixture of that area. In an area dominated by one class, it is nearly the accuracy of that class. The per-class accuracies are what answer for each one.

That the interval covers the error of MapBiomas. it covers only sampling variation: how many cells were compared. The error of the reference itself, the lag between its year and the window of the run, and the difference in definition between the two legends all stay outside it.

That a class with few cells has a measured accuracy. its interval is wide by construction. A class with a dozen reference cells can report $100\,\%$ producer accuracy with an interval that falls below $70\,\%$, and it is the interval that informs.

CHAPTER 7

Quantity and allocation: two errors that do not add up

Chapter 6 left a calculation half done. Overall accuracy says in what fraction of cells the two maps agree, and its complement is the disagreement. But the disagreement is not one thing.

Two runs can get the same fraction of cells wrong for opposite reasons, and the single number does not separate them. The application reports the separation.

7.1 The two parts

Let M be the matrix of the previous chapter, normalised by the total number of cells, so that its entries p_{ij} sum to 1. Two marginal sums matter. $p_{i\cdot}$ is the sum of row i, the fraction the classifier assigned to class i. $p_{\cdot i}$ is the sum of the column, the fraction the reference has of that class.

$$Q = \frac{1}{2}\sum_i |p_{i\cdot} - p_{\cdot i}| \tag{7.1}$$

Equation (7.1) is the **quantity disagreement**. For each class, it takes the difference between how much the classifier says exists and how much the reference says exists, in absolute value, and sums. The half avoids double counting: every excess in one class is a shortfall in another, and the sum of the signed deviations is always zero. Q therefore measures only how much the two *compositions* differ. It does not look at where anything is.

$$A = \sum_i \min(p_{i\cdot} - p_{ii},\ \ p_{\cdot i} - p_{ii}) \tag{7.2}$$

Equation (7.2) is the **allocation disagreement**. For each class, $p_{i\cdot} - p_{ii}$ is what the classifier called i and is not. $p_{\cdot i} - p_{ii}$ is what is i and it did not call so. The minimum of the two is the part that can be explained as a swap: a cell it gave to another class and received back in equal measure. What remains beyond the minimum is a difference in quantity, and has already been counted in Q.

The two parts close the calculation with the agreement:

$$\mathrm{OA} + Q + A = 1 \tag{7.3}$$

Equation (7.3) is an identity, not an approximation. The figure below checks it in every case it draws, before drawing.

7.2 The same disagreement, two opposite diagnoses

> The first bar is the case that justifies the decomposition existing. A classifier that swaps soybean for other temporary crops in equal measure **reproduces the composition of the reference exactly**, and errs in every cell it swapped. By composition, the two maps differ in nothing. By agreement, $30\,\%$ of the cells disagree. The quantity disagreement is the number that equals zero in that case, and that is why it cannot be the only one reported.

This is not a hypothetical case inside the product. The composition panel of the application compares the two marginal distributions, how much each map has of each class, and that

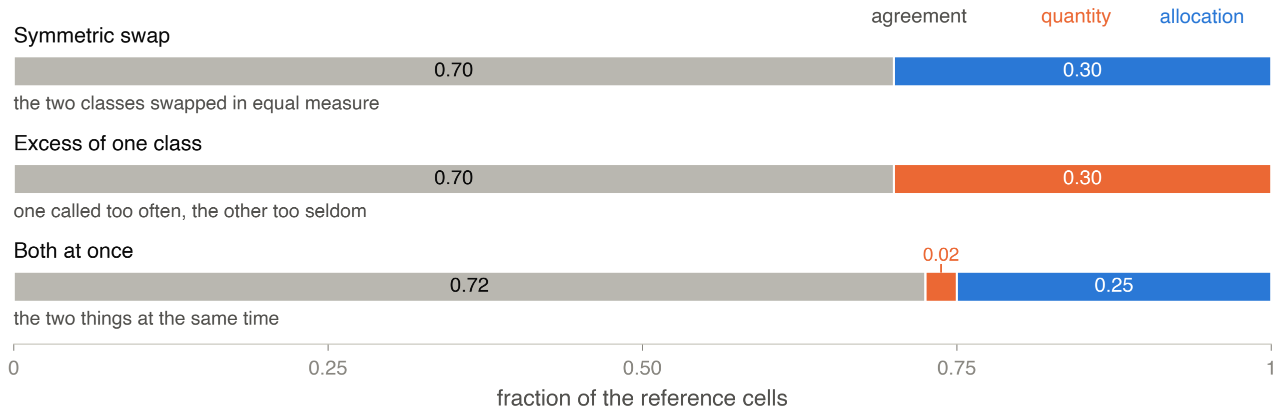

Figure 7.1: The decomposition of the disagreement over three synthetic example matrices. Each bar runs from 0 to 1 and represents the total of reference cells, split into three segments: **grey, the agreement**, the fraction in which the two maps say the same thing; **orange, the quantity disagreement**; **blue, the allocation disagreement**. The three segments of each bar sum to 1, which is the identity of equation (7.3). **First bar, symmetric swap:** the two classes are exchanged in equal measure, and the disagreement of 0.30 is entirely allocation, with quantity zero. **Second bar, excess of one class:** the classifier calls one class too often and the other too rarely, with no swap, and the same disagreement of 0.30 is entirely quantity. **Third bar, mixture:** disagreement of 0.28, split into 0.02 of quantity and 0.25 of allocation. The reading is the comparison between the first two: the total disagreement is identical, and the diagnosis is opposite. The matrices are built to separate the two regimes and are not runs of the product.

comparison is the quantity term on its own. The decomposition was added because the panel was showing half the calculation.

7.3 What each part calls for

The two parts point at different corrections, and that is where the separation pays.

Quantity high, allocation low. the two maps disagree about how much of each class exists, and agree about where. It usually comes from a threshold: the classifier is systematically generous or timid with one class. A comparison of area between the two maps becomes invalid; a comparison of spatial pattern still holds.

Allocation high, quantity low. the two maps agree about how much exists and disagree about where. Here the total area per class can be quoted with less risk, and the map does not serve to say which field is which.

Both high. there is no safe reading of the run over that area. It is the case the domain shift diagnosis exists to explain.

7.4 Kappa is not reported

The kappa coefficient would occupy this place in much of the remote sensing literature, and the application does not compute it. The omission is deliberate and is recorded in the code. The decomposition into quantity and allocation comes from the work that argues against kappa (Pontius and Millones, 2011), and it is what takes its place.

The argument, in summary, is that kappa corrects the agreement by an agreement *expected by chance* that depends on the marginals of the two maps. That correction mixes into one constant things the user needs to keep apart, and the resulting value is hard to interpret without going back to the matrix. Quantity and allocation are read directly, in the same unit as the accuracy, and sum with it.

7.5 What cannot be concluded

That the decomposition says which side the error is on. it splits the disagreement between two maps. Which of the two is wrong remains a field question, as chapter 6 already put it.

That quantity zero means the map is right. it means the two compositions match. The first bar of figure 7.1 has quantity zero and gets three cells in ten wrong.

That allocation zero means the places are right. it means there was no compensated swap. A classifier that calls one class too often everywhere has allocation zero and is wrong everywhere it overreached.

That the parts sum to an error rate. $Q + A$ is the total disagreement, which is $1 - \mathrm{OA}$. It is not an error rate of the classifier, for the reason of the previous chapter: part of the disagreement may belong to the reference.

CHAPTER 8

Agreement by block: where the disagreement is

The two previous chapters gave numbers that hold for the whole area. Overall accuracy says how much, and the decomposition says of what kind. Neither says *where*.

The difference matters because it separates two problems that call for opposite work. A uniformly mediocre classifier has a model problem. One that gets everything right except in one corner has usually met a landscape: a floodplain, the margin of a reservoir, a field of a crop the legend does not name.

8.1 The grid

The extent of the raster is divided into a fixed grid of six by six, which gives 36 blocks. Each reference cell is assigned to the block its first pixel falls in, and each block accumulates how many cells it received and in how many the two maps agreed.

Six per side is a compromise: it gives enough blocks to distinguish a corner from a centre, and keeps enough cells in most of them when the drawn area is small.

> A block with fewer than 20 reference cells has its count reported and its percentage omitted. The reason is in the code, and it is arithmetic: with three cells, the percentage can only read $0\,\%$, $33\,\%$, $67\,\%$ or $100\,\%$. Publishing it beside a block of several hundred would invite comparing the two as though they were the same measurement. The count travels regardless, so that it is visible why the percentage is missing.

8.2 What is reported, and why it is not a mean

The output is not the mean of the blocks. It is their distribution: how many blocks were measured, the median, the interquartile range, the minimum and the maximum.

The median and the quartiles, rather than the mean and the standard deviation, because the set is small and uneven. Thirty-six blocks, some out of the count by the floor, and a landscape under no obligation to distribute itself symmetrically. A mean with a standard deviation over that set would suggest a symmetry it does not have.

8.3 The same global number, two landscapes

The figure was made to show that the global number does not separate the two cases. On measuring it, it showed more than that, and the rest is about the statistics reported.

> In the right panel, the interquartile range is 1.7, practically equal to the 1.9 of the uniform panel. **The IQR does not see the low-agreement corner.** Four blocks in thirty-six fall outside the quartiles, so the measure of spread passes over them without registering them. And the **median is higher** in the case with the corner, 90.4 against 85.0. Read on its own, it ranks the run on the right above the other.
> What captures the corner is the **minimum**: 40 against 82. The distance between the median and the minimum is the reading, and not either of them alone.

The reading rule. compare the median with the minimum, and the minimum with the cell floor. A high median with a minimum far below it is the pattern of a localised landscape. A median

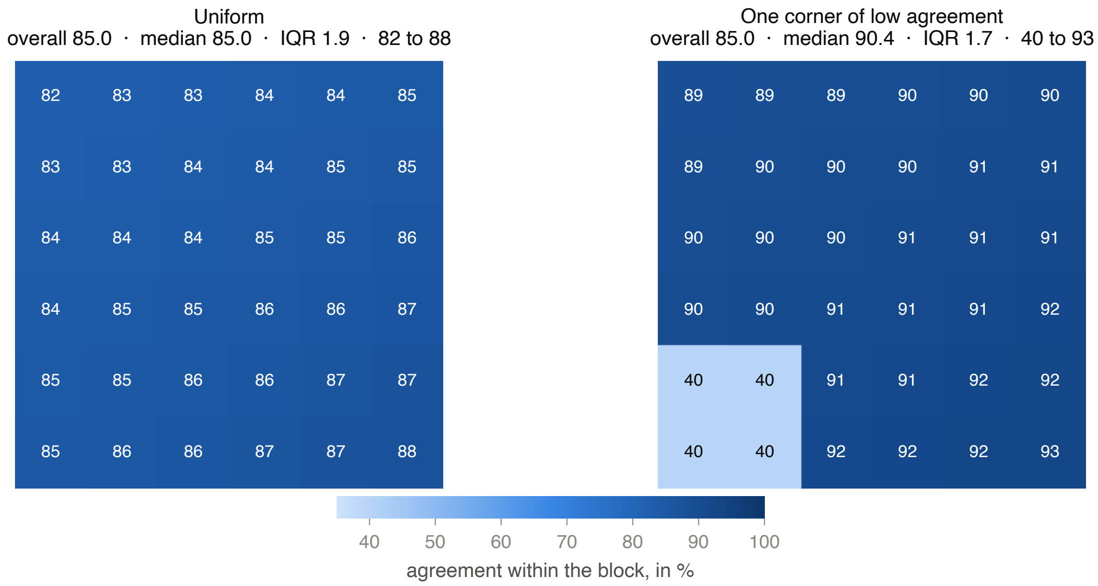

Figure 8.1: Two block grids with the *same* global agreement of 85.0 %. Each panel is the six-by-six grid over the extent of the raster, with the agreement of the block printed inside it and coded by colour on the scale in the footer, from light to dark as agreement rises. **Left panel, uniform:** every block between 82 and 88, median 85.0, interquartile range 1.9. **Right panel, one low-agreement corner:** thirty-two blocks between 89 and 93, and four blocks in the lower left corner at 40, visible as the pale block in the grid. Median 90.4, interquartile range 1.7, minimum 40. Both grids are built to separate the two regimes and are not runs of the product; the global agreement of the two is identical by construction.

and a minimum close together, with a narrow interquartile range, is the pattern of uniform performance, at any level.

8.4 It is not a test

The spread between blocks is descriptive, and the function that computes it declares as much. Three reasons, and none of them can be removed by adjusting the calculation.

The grid is fixed, not sampled. the blocks are a regular partition of the extent of the raster, not a sample designed to estimate anything.

The blocks do not have the same sample size. each one receives the cells that happened to fall in it, and the floor of 20 already removes some from the count.

They are not independent of the landscape. a field, a floodplain or an urban patch crosses several blocks, so neighbouring blocks err together by construction.

An accuracy assessment with a sampling design behind it is a different thing, and Olofsson et al. (2014) is the reference for it. What this chapter describes is not one.

8.5 What cannot be concluded

That the interquartile range measures the worst region. it does not. Figure 8.1 has two cases with almost equal IQR and minima that differ by 42 points.

That a high median ranks two runs. the median of the case with the low-agreement corner is higher than that of the uniform case. It describes the typical block, and the disagreement is not in the typical block.

That a block with no percentage is a low-agreement block. it is a block with fewer than 20 reference cells. Its count is in the payload, and the absence of the percentage is a refusal to report, not a low value.

That the difference between two blocks is significant. there is no test here, and the comparison between blocks has no sampling error attached. The same holds for comparing the spread between two areas.

CHAPTER 9

Class separability: contrast and signature

The previous chapters measured disagreement against a reference. This one measures something prior: whether the observed bands *can* distinguish two classes, independently of which model tries.

It is a property of the measured data, not of the classifier. A low distance says that those bands, in that acquisition, do not separate that pair. It does not say that a better model would separate them, and it does not point at a repair.

9.1 Where the numbers come from

The run stores, per predicted class and per band, the mean and the standard deviation of the reflectance. There are seven bands, from B02 to B12, and the record exists only for a class with at least 30 pixels in that band.

Two properties of that record decide how it is read.

It is one acquisition, not the series. the classification is temporal, with up to 22 dates, but the stored reflectance is that of the scene from the middle of the period. A mean of seven bands across a season would mix phenological stages into a curve that describes no date at all. The scene used is named in the payload.

It is physical reflectance, with a label from the other space. the spectrum is reported through `to_reflectance`, with the baseline 04.00 offset, because it is a physical quantity. The classifier that produced the labels consumed `as_trained`, without the offset. It is the seam of chapter 3: **the labels come from one space and the reflectance reported for them comes from the other**.

9.2 The distance, and why it saturates

Two classes summarised by a mean and a standard deviation in one band are two Gaussians, and two Gaussians have an overlap in closed form. The Bhattacharyya distance between them is

$$B = \frac{(\mu_1 - \mu_2)^2}{4\,(\sigma_1^2 + \sigma_2^2)} + \frac{1}{2}\,\ln\!\left(\frac{\sigma_1^2 + \sigma_2^2}{2\,\sigma_1\sigma_2}\right) \tag{9.1}$$

Equation (9.1) has two parts with distinct roles. The first is the separation between the means, divided by the joint dispersion: it grows when the classes move apart and shrinks when either of them spreads. The second is the disagreement between the dispersions on its own, and that is why two classes with the *same* mean and different standard deviations do not sit at zero.

B has no upper limit. Two pairs at $B = 8$ and $B = 14$ sit far apart on the scale while being equally separable in practice. The panel therefore reports the Jeffries-Matusita distance, which saturates:

$$\mathrm{JM} = 2\left(1 - e^{-B}\right) \tag{9.2}$$

Equation (9.2) takes $B \in [0, \infty)$ to $\mathrm{JM} \in [0, 2)$. Above about 1.9 the pair is separated and more distance changes nothing a classifier could use.

The conventional reading is the one the panel draws in bands. Below 1.0 the classes overlap substantially. Between 1.0 and 1.9 there is partial separation. Above 1.9 they are effectively separable in that band.

9.3 What the scale demands of the data

A distance on a scale of 0 to 2 does not itself say how far apart the classes have to be. Figure 9.1 converts the scale into standard deviations.

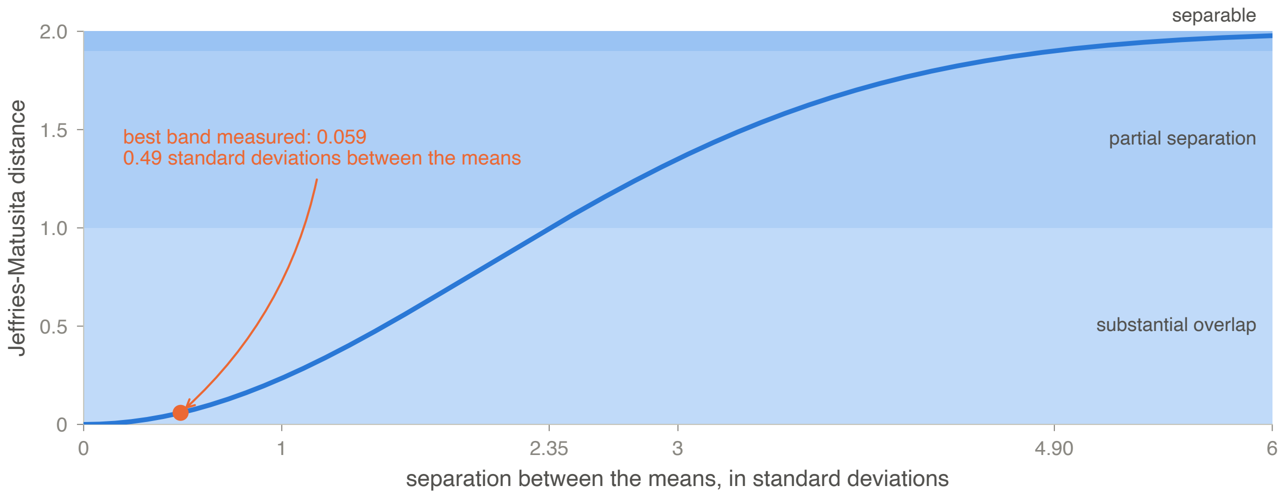

Figure 9.1: The Jeffries-Matusita distance as a function of how far apart the means of the two classes are, measured in standard deviations, for two Gaussians of equal dispersion. The horizontal axis is the separation between the means; the vertical is the distance, on the 0 to 2 scale the panel shows it on. The three horizontal bands are the reading bands of the panel, from lightest to darkest: **substantial overlap** below 1.0, **partial separation** between 1.0 and 1.9, **separable** above 1.9. The blue curve is equation (9.2), computed by the same function the application runs. The axis marks at 2.35 and 4.90 are the separations needed to reach the two band boundaries. **The orange point** is the best single-band value the line measured over a pair of regions, 0.059, and it corresponds to means 0.49 standard deviations apart.

The three numbers in the figure are the whole reading. For a band to leave substantial overlap, the means of the two classes have to be 2.35 standard deviations apart. To reach separable, 4.90. The best single-band value the line measured corresponds to 0.49 standard deviations.

9.4 The two readings, which do not move together

The same calculation answers two different questions, and the panel draws both because they can diverge.

Within one run, between two classes. how much one class stands out from the *others around it*, band by band. It is contrast against the surroundings.

Between two runs, for the same class. how far the signature of the class itself moved from one area to the other.

> A crop whose reflectance is almost the same in both regions can become unclassifiable in the second. What collapsed was not its signature: it was the contrast against everything else in the scene. Reporting only one of the two readings makes that case unreadable, and that is why the panel carries both.

9.5 The best of seven, and not the mean

When the panel summarises a pair of classes in one number, it uses the **maximum** over the seven bands, not the mean.

The reason is the question being asked. A mean over seven bands is pulled down by the six that do not carry that distinction, and reports as marginal a pair that one band separates well. What matters is whether *some* band separates the two, and which one.

9.6 What is refused, and why

A standard deviation equal to zero makes the second part of equation (9.1) diverge, and the implementation refuses the pair instead of regularising.

The current alternative is to add 10^{-9} to each variance. Under the conditions where that correction is usually applied, a continuous index over thousands of pixels, the degenerate case does not appear and the term only protects the logarithm. Here the standard deviation arrives rounded to six places, over at least 30 pixels, so it *can* be exactly zero. With 10^{-9}, that zero would become a distance near 2, and the most separable pair in the panel would be the one whose data was too coarse to be measured.

The output therefore distinguishes two reasons for absence. An *absent* band is one for which the class did not gather 30 pixels to measure. A *degenerate* one is a band whose standard deviation does not allow the calculation. In both cases the panel says which it was, instead of leaving a gap.

9.7 What cannot be concluded

That low separability condemns the classification. every distance here is univariate. A classifier that reads the bands jointly separates classes that no band separates on its own, and the difference between those two things is not small.

The reference figure for that measurement is the number in figure 9.1. A best single-band distance of 0.059, on a scale running to 2, over data on which a Random Forest reached a macro-F1 of 0.81.

That the measure holds for the season. it holds for one acquisition, that of the middle of the period. A crop can be inseparable on the date read and separable six weeks earlier.

That it measures the ability of a model. it measures the overlap of the observed distributions. It is not accuracy, it is not a prediction of accuracy, and it does not indicate which model path to choose.

That a high distance guarantees a correct answer. it guarantees that that band separates the two distributions summarised by mean and standard deviation. Real distributions are not Gaussian, and a summary by two moments ignores everything that is neither centre nor dispersion.

CHAPTER 10

Domain shift: the diagnosis

Chapter 4 left a problem open. The legend is closed, so an area outside what the model learned receives one of the five classes with confidence, and nothing on screen reports it.

The domain shift diagnosis is the partial answer. It compares two runs and measures how far the data of one departs from that of the other. It corrects nothing, and the next chapter deals with why not.

10.1 The fingerprint

Every classification stores a compact **fingerprint** of the data it consumed, and the comparison happens between two of those, without rereading a single scene.

The fingerprint holds a sample of up to 512 pixels of the feature vector, with the mean and the variance of each. It also carries an NDVI histogram of 32 fixed bins between -0.2 and 1.0, and the means of the red and the near infrared.

When the model path does not produce the 80-feature vector, an NDVI-only fingerprint is stored. The measures that depend on the full vector degrade rather than disappear.

Standardised against training, not against the run. the comparable values come in standard deviations of the *training* set, not of the current run. That is what gives sense to the phrase "*this feature is three deviations out*". They are three deviations from the centre of what the forest was fitted on, not three deviations from itself.

10.2 The four measures

Table 10.1: The measures the diagnosis reports, with what each one sees and what it does not reach.

Measure	What it compares	What stays outside
KL on NDVI	the two NDVI distributions, bin by bin	NDVI only; nothing of the other bands nor of the temporal shape
CVA	the distance between the two mean vectors, in training deviations	everything that leaves the means equal
MMD with a Gaussian kernel	the two whole distributions, not just the centre	power falls with dimension; 80 features over 512 rows is few
Per-feature table	which feature moved, and how much, in training deviations	the importance that weights it orders, it does not quantify

KL. is computed in both directions and reported symmetrised, as the mean of $\mathrm{KL}(A\|B)$ and $\mathrm{KL}(B\|A)$, with both directions also available. The divergence is not symmetric by nature, and reporting one direction alone would make the reading depend on which run was called the first.

CVA. is the norm of the difference between the two mean vectors. Because the units are training deviations, the number reads directly. A CVA of 3 says that the centre of run B is three deviations from the centre of A, on the training scale.

The direction of that change is also reported, as the angle of the vector in the red-near infrared plane. Seasonality and deforestation pull towards different sides of that plane.

10.3 Why the MMD kernel is Gaussian

The MMD compares two whole distributions through a kernel, and the choice of kernel decides what it can see.

With a linear kernel, the statistic reduces to $\|\mu_A - \mu_B\|$, which is the magnitude of the change vector computed another way. In real runs the two agreed to the last digit reported, so the report carried one measure under two names.

Worse than the redundancy is the blindness. Figure 10.1 measures it.

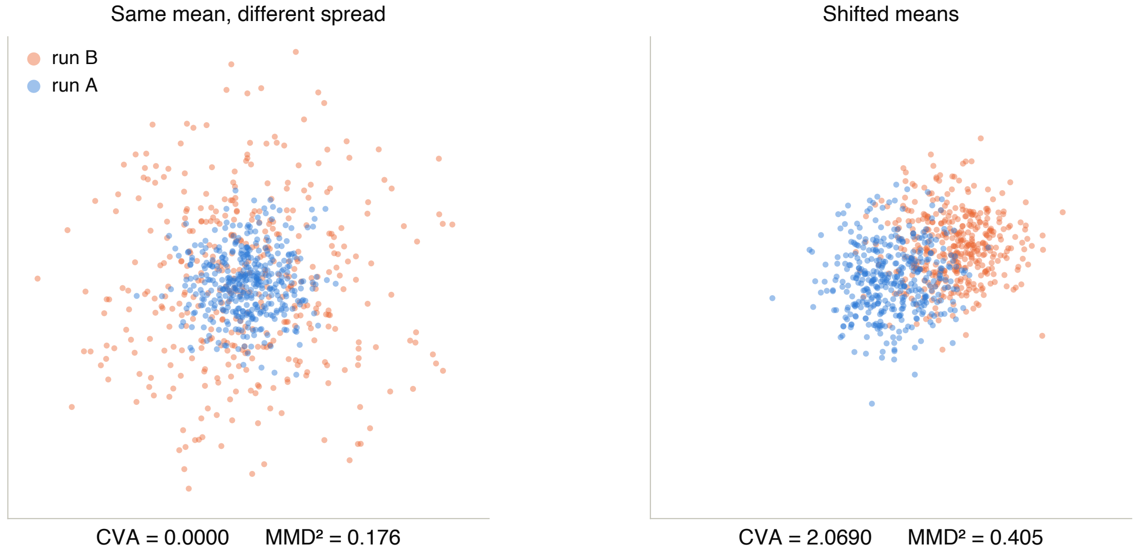

Figure 10.1: Two comparisons between samples, with both measures computed by the functions of the sidecar itself. Each panel shows two point clouds, run A in blue and run B in orange, and carries below them the magnitude of the change vector and the squared MMD. **Left panel:** the two clouds have the same mean and different dispersions, with that of B more than twice that of A. The magnitude of the change vector is $0.000{,}0$, and the squared MMD is 0.176: the first measure sees no difference at all, and the second sees one. **Right panel:** the clouds have the same dispersion and displaced means. The magnitude is $2.069{,}0$ and the squared MMD is 0.405, and both measures register the difference. The samples are synthetic and two-dimensional so that they fit on the page; the fingerprint of a run has 80 dimensions.

The left panel is the shape of conditional shift: the *same* class emitting a different spectrum, with a similar centre. A measure that looks only at the centre gives zero for it. That is the reason the kernel is Gaussian, which is characteristic: the MMD is zero only when the two distributions are equal (Gretton et al., 2012).

The MMD estimator is unbiased, and can therefore come out slightly negative when the two samples come from the same distribution. The negative value is reported as it is, without clipping at zero. A small negative *is* the sign that the two are indistinguishable at that sample size, and clipping it would erase that information.

10.4 The per-feature table

The three previous measures say *that* the domains differ. The per-feature table says *where*.

It lists the shift of each feature between the two runs, in training deviations. Beside it comes a column that multiplies that shift by the importance the forest assigns to the feature. A large movement in a feature the model barely separates on is not the same thing as a small movement in one it depends on.

The importance used is the impurity one, which is biased in favour of features with many distinct values (Strobl et al., 2007). It serves to **order**, not to quantify. The two columns are reported side by side, and neither is presented as the answer on its own.

Beside that, the two-dimensional projection of the two sample sets allows the separation to be seen rather than read. It is descriptive, and the position of a point in it has no unit.

10.5 What cannot be concluded

That a small MMD means the absence of shift. the power of the statistic falls with dimension (Ramdas et al., 2015), and 80 features against 512 rows is not the regime in which a small value authorises that conclusion. The module itself records this.

That the diagnosis corrects anything. it measures. Domain adaptation is declared outside the scope of the module, and the next chapter says why.

That a low KL means the domains are close. the KL here is computed over NDVI and nothing else. Two areas can have the same NDVI distribution and differ in every other band, or in the shape of the curve across the season.

That a feature at the top of the table is the cause. the table orders shifts. Correlated features move together, and the ordering by impurity importance carries the bias of the note above.

That the comparison is against the training set. it compares two *runs*. The standardisation is done in training deviations, which makes the numbers legible on that scale, but neither of the two runs is the training set.

CHAPTER 11

Domain shift: what it does not correct

The previous chapter described a diagnosis that measures and does not repair. Domain adaptation is declared outside the scope of the module, and that usually sounds like pending work.

It is not. The line tested adaptation, and its result is the reason the application does not ship it.

11.1 Six corrections, none of which worked

The problem has a constraint that is not a simplification: there are abundant labels in the source region and none in the target, so fine tuning is excluded by construction. The corrections have to be blind to the target label.

The line tested six families over a real pair, Iowa against western Paraná. Same sensor, same crop, opposite hemispheres. They range from pure calendar to a network that receives climate and learns what to do with it.

> None of them worked. The transfer gap stays around 40 percentage points in all of them, and the best method beats the baseline by 0.5 point, inside the spread between seeds. The ceiling in the source domain is 0.996, so this is not a weak classifier.

11.2 The physical correction wins in one place and loses in the other

One of the six corrections has a physical basis rather than a statistical one. It reindexes the time series in accumulated growing degree days, instead of calendar days, to compare the two regions in development time and not in clock time.

Figure 11.1 shows what it did in the two experiments of the line.

The inversion calls for an explanation, and it is not that one of the two experiments is wrong.

In the synthetic case, the thermal phase was planted. the generator built the shift with sowing date and thermal accumulation among the mechanisms. A phase correction resolves, by definition, a shift whose cause is phase.

In the real case, the pair crosses the equator. the literature reporting a gain from thermal time, of 21.2 points, measures a temperate pair against a temperate pair, in the same hemisphere, with no season inversion. Between Iowa and Paraná the window of thermal overlap covers only part of the cycle on each side.

> The conclusion that survives both is neither "*degree days work*" nor "*degree days do not work*". It is that reindexing in thermal time corrects the shift *when the shift is one of thermal phase*, and does not correct it when the gap is of another nature. Applying the correction requires a prior diagnosis saying which of the two cases one is in, and that diagnosis does not exist.

11.3 Two traps measured along the way

The corrections do not commute. in the synthetic experiment, shape normalisation applied on its own costs 12.3 points, and applied over landmark registration it yields 14.3. The order is to align time first and radiometry afterwards. An ablation reporting corrections in isolation measures something else.

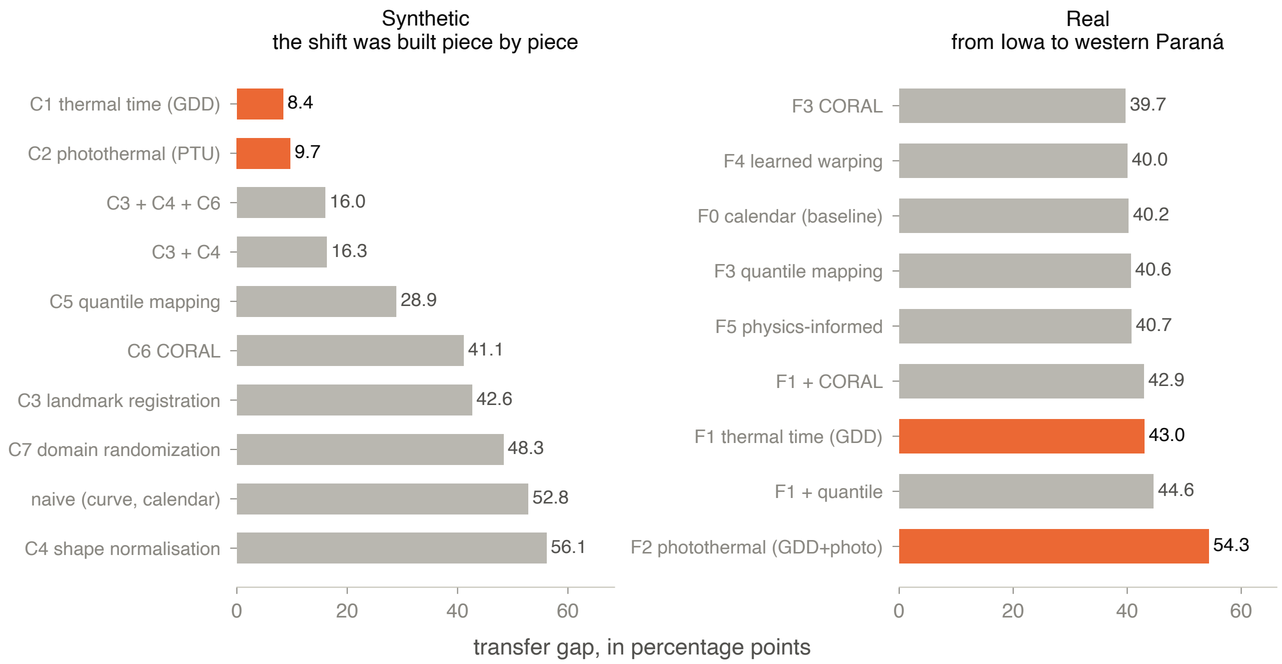

Figure 11.1: The transfer gap of each correction family, in two experiments. A shorter bar is better: the gap is how far performance on the target falls below the ceiling on the source, in percentage points. **Left panel, synthetic experiment:** the shift between the two regions was built piece by piece, with sowing, temperature, photoperiod, soil and cloud as named mechanisms. The gap without correction is 52.8, and reindexing in degree days takes it to 8.4, the smallest of all. **Right panel, real pair from Iowa to western Paraná:** every family sits between 39.7 and 54.3, and the pure calendar baseline is at 40.2. **The orange bars are the thermal time corrections**, the same ones in both panels. On the left they occupy the first two positions; on the right, reindexing in degree days gives 43.0, worse than correcting nothing at all, and the photothermal version gives 54.3, the worst of all.

No label-free indicator chose the method. the line tested three candidates to stand in for the label and rank the methods, and all of them failed. The entropy of the prediction correlates with real performance at $+0.29$, with the *sign inverted*: the model is most confident exactly where it errs most.

11.4 What reinterpreted all the rest

The studies above compared against MapBiomas on the target, and chapter 6 has already recorded that MapBiomas is a map produced by another classifier. A low ceiling on the target could be physical overlap between the classes **or** error in the reference, and none of those designs separated the two.

A later study separated them. It swaps the pair for two domains that have producer declaration labels on both sides, keeping the climatic contrast.

> The ceiling on the target is 0.99, not 0.65. With the same NDVI and the same classifier, what had looked like a physical limit of separability was, in large part, **error in the reference**. That retroactively reinterprets the numbers in this chapter: a large fraction of the roughly 34 points called irreducible was MapBiomas error, and not overlap between soybean and pasture.

For whoever uses the application, the consequence joins this chapter with chapter 6. Suppose the domain diagnosis reports a distance and the agreement falls. There are two possible causes, and the product does not separate them. The area may be outside what the model learned, or the reference it is measured against may be wrong there.

11.5 What cannot be concluded

That domain adaptation does not work. it did not work on that pair, with those six families, against that reference. The last clause is the one the material itself overturned afterwards.

That the 40-point gap is physical. much of it was label error in the reference, measured by contrast with a pair that had declaration labels.

That thermal time should be avoided. it recovers 44.8 points when the shift is one of thermal phase. What is missing is the diagnosis that says when that is the case.

That the confidence of the model helps to choose. the measured correlation between prediction entropy and real performance has an inverted sign. On that pair, it would lead to the wrong choice.

Part IV

Water and terrain

CHAPTER 12

Surface water by spectral index

Surface water is the product that inherits least from the limitations of the previous chapters. There is no trained model, so there is no fixed legend and no domain to leave.

What there is are three published indices, one threshold per date, and two implementation decisions that decide whether the result measures water or measures the absence of an observation.

> This product has no dedicated study in the *TERRA-Research* line. The three indices are published and cited below, and the implementation was ported from an earlier analysis in the same project. Chapter 1 puts that in the validation table: what supports the number here is the original source of each index, not a measurement of our own.

12.1 The three indices

All of them start from the green, B03, and differ in what they put on the other side of the difference.

$$\mathrm{NDWI} = \frac{\rho_{\mathrm{B03}} - \rho_{\mathrm{B8A}}}{\rho_{\mathrm{B03}} + \rho_{\mathrm{B8A}}} \tag{12.1}$$

Equation (12.1) contrasts the green with the near infrared (McFeeters, 1996). Water absorbs strongly in the near infrared and reflects relatively well in the green, so the difference is positive over water and negative over vegetation, which does the opposite.

$$\mathrm{MNDWI} = \frac{\rho_{\mathrm{B03}} - \rho_{\mathrm{B11}}}{\rho_{\mathrm{B03}} + \rho_{\mathrm{B11}}} \tag{12.2}$$

Equation (12.2) swaps the near infrared for the shortwave one (Xu, 2006). Built surfaces are also dark in the near infrared, and NDWI confuses them with water. In the shortwave infrared, water absorbs even more and concrete does not. It is the index the application uses by default.

$$\mathrm{AWEI} = 4\,(\rho_{\mathrm{B03}} - \rho_{\mathrm{B11}}) - (0.25\,\rho_{\mathrm{B8A}} + 2.75\,\rho_{\mathrm{B12}}) \tag{12.3}$$

Equation (12.3) abandons the normalised form and uses fitted coefficients (Feyisa et al., 2014). The first term is the MNDWI contrast multiplied by four; the second discounts reflectance in the two infrareds, which is where dark soil and shadow still respond. The result is not bounded between -1 and 1.

> The variant implemented is **$\mathbf{AWEI_{nsh}}$**, the *no shadow* one. The original work proposes two, and the other is the one that treats shadow explicitly. None of the three indices in this chapter distinguishes water from cloud shadow or hillside shadow.

B8A and not B08. where the original formula asks for the near infrared, the implementation uses the narrow band B8A, at $20\,\mathrm{m}$, and not B08 at $10\,\mathrm{m}$. The choice comes from the analysis the code originated in, whose stack was assembled over the Prithvi band set. Using B08 would change NDWI and AWEI and would not reproduce the reference series.

12.2 The threshold per date

The literature cut is zero for all three indices. The application also estimates a threshold per date by the Otsu method, which picks the cut that maximises the variance between the two classes of the histogram.

There is a detail about that threshold that has to be read together with the result.

> The Otsu threshold is bounded to an empirical interval between -0.2 and 0.4, so that it does not run to a degenerate cut when the area is almost all land. That lower bound is reached often. In the reference holdings it saturates on 12 of 22 dates, 9 of 21 and 20 of 20. When that happens, the reported value is the *bound*, not an estimate, and the output marks it as such.

12.3 The two corrections that decide the product

The implementation diverges deliberately from the analysis it came from, at two points. Both deal with the same confusion: the absence of an observation read as the absence of water, or as its presence.

Validity per date, and not by the union of the dates

The prototype built a single validity mask, by the union of the dates. A pixel missing on one date still counted on that date. Its bands are zero, so every ratio index evaluates to exactly $(0-0)/(0+0+\varepsilon) = 0.0$. The comparison is index ≥ 0.0, so **the pixel enters as water**.

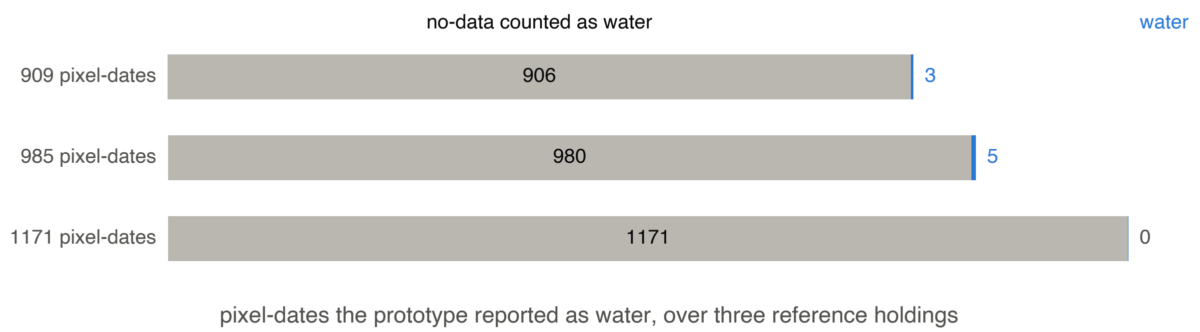

Figure 12.1: What the prototype counted as water, in three reference holdings. Each bar is the total of pixel-date pairs it reported as water at the fixed threshold, and the number on the left gives that total. **The grey part is the fraction of those pairs in which the pixel had no observation on that date** and entered as water by arithmetic, with its bands at zero. **The blue part, on the right, is what was left of actual water**, with the count printed beside it. In the three holdings: 906 of 909, 980 of 985 and $1{,}171$ of $1{,}171$. In the third, the blue bar does not exist: none of the reported pairs was water.

> Carrying that behaviour over would have produced a water product that measures missing observations. Validity is now evaluated per date, from the bands the selected index consumes. The final mask is intersected with it again, so that an unobserved pixel can never be reported as water.

Denominator per pixel, and not by the total number of dates

The second correction is of the same family. The occurrence map is the fraction of dates on which a pixel was water, and the prototype divided by the total number of dates in the window. That mixes dry with absent. A pixel seen on four dates and wet on one reads 0.25 under the correct denominator, and much less than that if the denominator is the whole window.

The denominator is now per pixel. A pixel never observed returns an absence of value, not zero.

12.4 Persistent and ephemeral

Over the occurrence, the application separates three bands:

Table 12.1: The occurrence bands, with the fraction of observed dates on which the pixel appeared as water.

Band	Occurrence	How to read it
Persistent	above $70\,\%$	water on most of the dates the pixel was seen
Ephemeral	$15\,\%$ to $70\,\%$	water on some of the dates; reservoir margin, floodplain, irrigation sheet
Below the band	less than $15\,\%$	not reported as water

The first two bands are reported separately and never summed. Summing them would produce a water area that exists on no date at all: the maximum extent reached at some point in the window, presented as though it were one water surface.

12.5 What cannot be concluded

That the mask distinguishes water from shadow. it does not. The AWEI variant implemented is the no-shadow one, and none of the three indices treats cloud or hillside shadow.

That the sum of persistent and ephemeral is the area of the water body. it is the union of two things measured on different dates. The area of a water surface on one date is the mask of that date.

That the reported threshold is always an estimate. when it hits the empirical bound, the value is the bound. The output marks this, and in the reference holdings it happened on most dates in two of the three cases.

That the occurrence is comparable between pixels. the denominator is per pixel, and two pixels with an occurrence of $50\,\%$ may have been seen on twenty dates and on two. The observation count travels with the value.

That the product inherits the limitations of the classifier. it does not. There is no trained model here, so there is no fixed legend and no training domain, and chapters 4 and 10 do not apply to it.

CHAPTER 13

The terrain chain: from the grid to HAND

The flood envelope does not read Sentinel-2. It reads elevation models and runs a classical terrain analysis chain over them. What comes out of it is a height: how far each cell sits above the drainage it flows to.

This chapter describes that chain and the decisions inside it that change the result. The product that comes out of it is the subject of the next chapter.

13.1 The chain, step by step

The sequence is that of Nobre et al. (2011), in five steps. Fill the depressions, direct the flow, accumulate it, extract the drainage by an area threshold, and measure the height of each cell above the drainage it flows to.

Filling the depressions. uses Priority-Flood with an epsilon (Barnes et al., 2014). The result is a model with no interior minimum and no plateau: every cell ends at least ε above the one the water left it through. The epsilon is not cosmetic. Without it, a filled lake becomes a plateau, and over a plateau there is no defined flow direction.

Directing the flow. each cell points at the neighbour of steepest descent among the eight, and a cell with no lower neighbour receives -1: there the water leaves the domain. After filling, that happens only at the edge, and it is that property which makes the graph acyclic.

Accumulating. counts how many cells drain through each one, itself included, following the topological order of the graph.

Extracting the drainage. a cell is drainage when its contributing area passes a threshold. The default is $0.5\,\mathrm{km}^2$, which at $30\,\mathrm{m}$ on a side is about 555 cells.

Measuring the height. walks the topological order backwards, from downstream to upstream, so that the drainage elevation of the receiver is already resolved when the cell is visited.

> The height is measured against the **original DEM**, not against the filled one. Inside a filled depression that produces negative HAND, and the value is information: the cell is *below* its drainage. The consumer decides whether to clip at zero, and the module does not clip.

13.2 Why the chain is reimplemented

There are ready-made libraries for this, and the application does not use them. There are two reasons, and both are declared in the code.

Compiled dependencies. the usual libraries bring extensions that have to be built. The TERRA distribution builds its own interpreter, and a package that fails to compile there fails at installation, on a machine with no compiler and nobody to read the error.

The drainage threshold. is a free parameter that almost no flood study reports, and this product measures its effect. A measurement of a parameter buried inside a dependency is a measurement of something the reader does not see.

The cost of that choice is measured. A grid of 900 by 900, which at $30\,\mathrm{m}$ covers a square $27\,\mathrm{km}$ on a side, takes $3.5\,\mathrm{s}$ for the whole chain. That is about $4.3\,\mu\mathrm{s}$ per cell.

13.3 Four products, one grid

The envelope compares elevation products, so each of them has to be read over the same ground, at the same moment and by the same path. Otherwise a difference between two products is a difference between two ways of reading.

There are four: `cop-dem-glo-30`, `nasadem` and `alos-dem`, all at $30\,\mathrm{m}$, and `cop-dem-glo-90` at $90\,\mathrm{m}$. All of them come from the same catalogue the application already reads for imagery.

13.4 Why the window is larger than the area

The HAND of a cell depends on the area that drains to it, and that area can come from outside the drawn region. The read is therefore done over a window 2 to $5\,\mathrm{km}$ beyond the area, with every intersected tile merged.

The catalogue search is done over the **enlarged window**, not over the area. A tile that only touches the margin ring would never be returned by a search over the area, and the merge would be left with a hole exactly where the water enters.

Measured over Copernicus GLO-30 in the lower Taquari, cutting the model at the edge of an area $8\,\mathrm{km}$ on a side costs 0.4 percentage point of extent. Between 2 and $4\,\mathrm{km}$ it costs 1.9 to 2.4 points. The effect exists, and it is modest for an area of several kilometres.

> Below about $1.5\,\mathrm{km}$ on a side it **inverts sign**: the clip starts reporting *more* extent than the correct read, $21.0\,\%$ against $14.5\,\%$.
> The cause is arithmetic. The cell of that model measures $827\,\mathrm{m}^2$, so the drainage threshold of $0.5\,\mathrm{km}^2$ is 605 cells. An area $1\,\mathrm{km}$ on a side contains 1369 cells in total, and the maximum contributing area any cell in it can accumulate is $1.13\,\mathrm{km}^2$. Only 16 cells cross the threshold, all of them at the outlet of the clip. With no internal drainage network, almost every cell measures its height against the edge the water leaves by, which is close and low: HAND collapses and the extent inflates.

The rule that follows. in an area whose side is of the order of the square root of the drainage threshold, the margin is not a refinement. The value reported without it is not an imprecise estimate of the extent: it is a different quantity.

13.5 What the chain produces

13.6 The drainage threshold decides the result

The paragraph above named the drainage threshold as the reason the chain is reimplemented. Table 13.1 measures why, over the same terrain as the figure.

Table 13.1: The effect of the drainage area threshold on the reported extent, in the lower Taquari, with the sidecar chain. The HAND threshold is the same in every row; what changes is how many cells count as drainage.

Threshold	In cells	Drainage cells	Extent at $1\,\mathrm{m}$	Extent at $2\,\mathrm{m}$
$0.1\,\mathrm{km}^2$	121	$5.26\,\%$	$14.2\,\%$	$18.8\,\%$
$0.5\,\mathrm{km}^2$	**605**	$2.49\,\%$	$10.2\,\%$	$13.5\,\%$
$2\,\mathrm{km}^2$	2,419	$1.32\,\%$	$7.6\,\%$	$10.0\,\%$
$10\,\mathrm{km}^2$	12,096	$0.63\,\%$	$4.9\,\%$	$6.3\,\%$
$50\,\mathrm{km}^2$	60,480	$0.36\,\%$	$3.3\,\%$	$4.2\,\%$

The row in bold is the application default. Between the first row and the last, the extent at

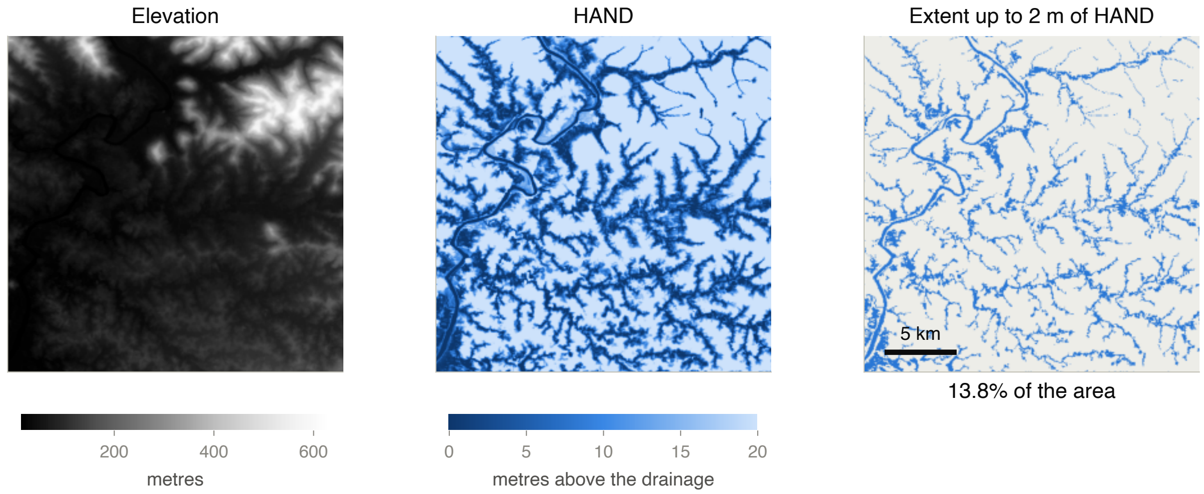

Figure 13.1: The three layers of the chain over the same real place: the lower Taquari valley, in Rio Grande do Sul, in a square about 23 km on a side containing Lajeado, Estrela and Cruzeiro do Sul. **Left panel, elevation:** Copernicus GLO-30 as the chain receives it, from dark (low) to light (high), with the river meandering along the valley floor and the slopes around it. **Middle panel, HAND:** the height of each cell above the drainage it flows to, in metres, saturated at 20. Dark blue marks what is close to the drainage, and the dendritic network that appears is the valley structure itself, not a drawn river. **Right panel, the extent up to 2 m of HAND:** the cells below that threshold, in blue, over the higher terrain in grey. They are 13.8 % of the area, and their shape is the valley floor of the whole network, not only the main channel. The chain is the one in the sidecar, with the drainage threshold at the default of $0.5\,\mathrm{km}^2$.

2 m goes from 18.8 % to 4.2 %. That is **a factor of four and a half**, with neither the terrain nor the HAND threshold having changed.

> Two HAND extents are comparable only if they were produced with the same drainage threshold. An extent figure without that parameter declared is not verifiable. That is why the application keeps it fixed, at $0.5\,\mathrm{km}^2$, and declares it in the result.

> The solar chain does not do this. Chapter 2 has already recorded it: `solar.fetch_dem` reads the first tile the catalogue returns, with no merge. Copernicus tiles are one degree, so an area crossing a border receives a model that covers part of it. For slope and aspect this degrades in patches, and it is a limitation of the irradiation-over-terrain and site maps, not of this chain.

13.7 What cannot be concluded

That a HAND extent is a flood extent. HAND is a height above the drainage. There is no rain, no discharge, no routing and no time. A HAND threshold delimits the terrain that would sit below a water sheet of that height, not the sheet an event would produce.

That the drainage area threshold is neutral. it decides which cells are drainage, and therefore what all the others measure their height against. The default of $0.5\,\mathrm{km}^2$ is the choice of the reference, and the product keeps it fixed and declared rather than sweeping it.

That negative HAND is an error. it is a cell below its drainage, inside a depression the filling closed. The module does not clip at zero.

That the margin resolves the truncation. it pushes it further away. The contributing area of a floodplain cell is unbounded in principle, and what comes from beyond the margin is still missing.

That the measured numbers hold for another basin. figure 13.1 and table 13.1 measure one specific floodplain. What transfers is the shape of the argument and the cell arithmetic, not the percentage points.

CHAPTER 14

The envelope: the agreement raster

The previous chapter ended at an extent: the cells below a HAND threshold. What is missing is the question that decides whether it can be used.

A HAND extent comes from an elevation model, and there are several of those over the same ground. If two products disagree about where the drainage is, they will disagree about the extent. The TERRA product is not the extent of one model: it is the extent *with the divergence between models beside it*.

14.1 The agreement raster

Four elevation products are read over the same grid, and the chain of chapter 13 runs once for each. At the reference threshold of $1\,\mathrm{m}$, each one produces its mask.

The agreement raster counts, cell by cell, how many of them call that cell flooded. The number runs from 0 to 4, and the reading is direct.

> Where the count is **0 or 4**, the products do not disagree: the terrain decided. Where it falls **between 1 and 3**, the choice of elevation product decided.
> A single mask with an accuracy beside it does not show this. It gives a shape, and the shape changes with the product without anything on screen saying so.

In that run, of the $20.5\,\mathrm{km}^2$ that some product calls flooded inside the area, $4.9\,\mathrm{km}^2$ are unanimous and $15.6\,\mathrm{km}^2$ are contested. **That is** $76.1\,\%$ **of the wet extent decided by the choice of elevation model**, and not by the terrain.

> A count of zero is not a level of agreement. A cell no product calls flooded carries no measure of accord between them: it is the dry remainder of the area, and it is reported as such. The contested fraction is taken over the wet extent, not over the whole area, because it is of that extent that it is a part.

14.2 IoU, which is the CSI of the literature

The agreement between two products is measured by the Jaccard index between the two masks:

$$\mathrm{IoU} = \frac{|A \cap B|}{|A \cup B|} \tag{14.1}$$

In equation (14.1), A and B are the sets of cells each product calls flooded. For binary masks that value is numerically the Critical Success Index of the flood literature, $\mathrm{TP}/(\mathrm{TP}+\mathrm{FP}+\mathrm{FN})$. The same column reads as CSI, with no conversion.

That equivalence opens a reading about the reach of the method.

> When a HAND method is evaluated against an observed extent, the CSI it reaches is compared against that of other methods. The line in figure 14.1 at $1\,\mathrm{m}$ is **the CSI the baseline scores against itself**, measured between elevation products of the same method. It is a ceiling on what any HAND method can reach in that area with that data.

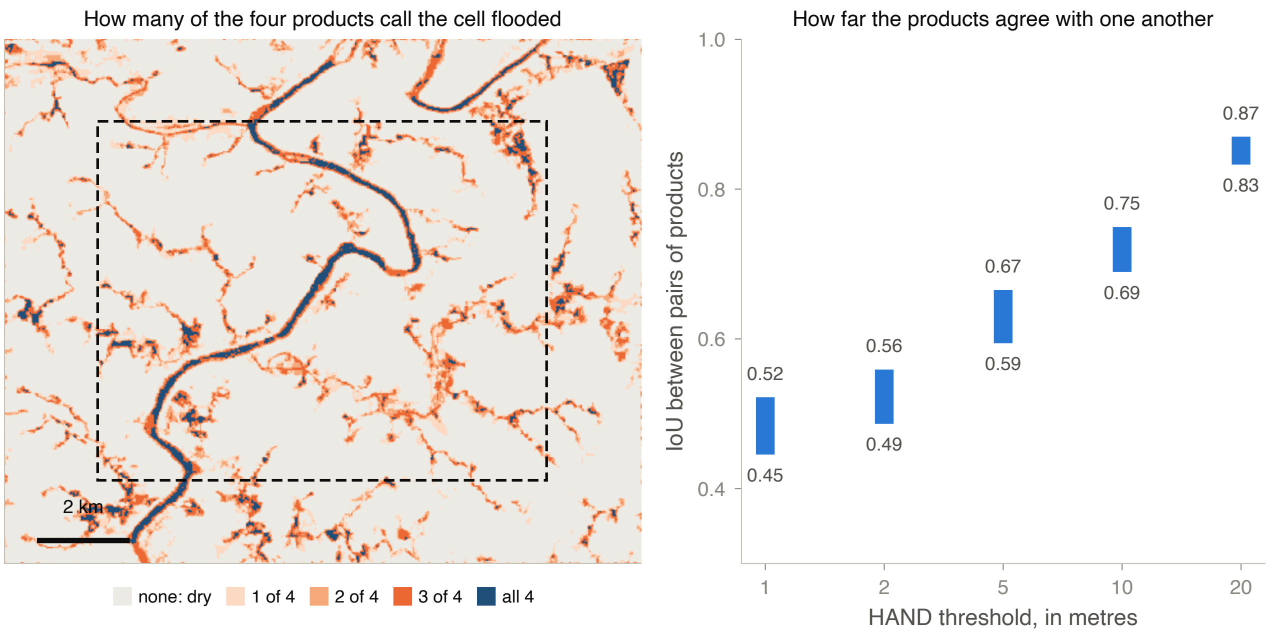

Figure 14.1: A run of the envelope over a clip about 8 km on a side in the lower Taquari, between Lajeado and Estrela. **Left panel, the agreement raster at the reference threshold of 1 m:** each cell takes the colour of the count of products calling it flooded, from grey (none of the four) to dark blue (all four), with the three shades of orange in between. The dashed rectangle is the drawn area, and what lies outside it is the margin the chain reads and does not report. The reading is the shape: **the main channel appears in blue, continuous**, and around it there is a wide fringe of orange. The terrain decides the channel; the fringe is decided by the choice of model. **Right panel, the agreement between pairs:** for each HAND threshold, the bar runs from the smallest to the largest IoU among the six pairs of products. At 1 m the pairs sit between 0.45 and 0.52; at 20 m, between 0.83 and 0.87. The tighter the threshold, the less the products agree.

14.3 The disagreement is not an edge effect

Products can disagree through an edge effect: clipping, resampling, drainage truncation. The product tests that hypothesis instead of dismissing it.

Beside each IoU comes an **inset IoU**, computed over the area shrunk by a ring of $1\,\mathrm{km}$ cut from its own edge. If the disagreement were an edge effect, the inset value would rise.

In the run in the figure it does not rise. At $1\,\mathrm{m}$, the range over the area is 0.45 to 0.52, and over the inset it is 0.42 to 0.52.

The ring has a ceiling. in a small area, a ring of $1\,\mathrm{km}$ would consume the whole polygon. It is capped at half the shorter side of the area, and the value actually used is reported together with the inset IoU. Without that, an inset of $1\,\mathrm{km}$ would be compared with one of $200\,\mathrm{m}$.

14.4 What the output declares alongside the numbers

Three parameters accompany the measurement, and without them none of the numbers is comparable.

The drainage threshold. fixed at $0.5\,\mathrm{km}^2$, for the reason table 13.1 measures: it moves the extent by a factor of four and a half.

The reference threshold of the raster. $1\,\mathrm{m}$. It is where the study that originated the product reports the largest disagreement between models. An envelope drawn at a higher threshold reports a smaller disagreement, as the right panel of the figure shows.

The area against the window. every count, area and fraction in the payload is taken inside the drawn polygon. The window the chain ran over is larger, and the two appear side by side in the result. Summarising the window instead of the polygon would inflate every area figure.

> The model set of TERRA is not that of the study the product originated in. That study used four products obtained with credentials from another catalogue, and this one reads the catalogue the application already reads, without credentials. That catalogue has no SRTMGL1, and `alos-dem` takes its place. **The range published by the study is not a prediction of the range TERRA measures**, and the result carries that qualifier in the payload itself.

14.5 What cannot be concluded

That the contested fringe is error. it is the interval within which the available models do not decide. None of the four is ground truth, and a contested cell may be flooded or dry.

That a high IoU means the extent is correct. it means the products agree with each other. Four models inheriting the same bias agree without any of them being right.

That the contested fraction of one area holds for another. it depends on the relief, the threshold and the set of products. In the run in the figure it is $76.1\,\%$ of the wet extent; over different relief it will be a different number.

That the envelope covers the uncertainty of the product. it covers one source of it, the choice of elevation model. The drainage threshold, the HAND threshold and the absence of rain and discharge remain outside, and all three already appeared in chapter 13.

Part V

Energy

CHAPTER 15

Solar resource at a point

The previous products depend on a scene. This one does not. It is a physical chain over reanalysis series, with no trained model, no fixed legend and no training domain to leave. Any area on Earth returns an answer, and no answer fails for want of an image.

What changes with that is the question to ask. It is not whether the model got that pixel right. It is at what scale the number refers, and against what it was checked.

15.1 The cell that answers for the area

Chapter 2 recorded the two NASA POWER grids: radiation from SYN1DEG on a one-degree cell, meteorology from MERRA-2 on a $0.5°$ by $0.625°$ cell. What is missing is the measurement of what that scale costs.

The line measured over three holdings in western Paraná. Two of them are about $3\,\mathrm{km}$ apart and the third is $70\,\mathrm{km}$ to the south. The series of the three points were compared day by day over $10{,}958$ days.

> The **global horizontal irradiance** (GHI) and the **direct normal irradiance** (DNI) are **identical** at the three points. The maximum difference is 0.00, and not a small difference: the three fall in the same one-degree cell.
> The air temperature differs by up to $3.06\,°\mathrm{C}$ and the wind by up to $1.55\,\mathrm{m\,s^{-1}}$. The meteorological grid is finer than the radiation grid, and it separates the third holding from the other two.

Comparing the solar resource of two holding-scale areas, with this source, returns zero by construction. What the answer describes is the cell the area fell in.

What the lack of resolution costs. the Global Solar Atlas resolves irradiance at $250\,\mathrm{m}$. Over the same three points, the GHI spread it sees is $15\,\mathrm{kWh\,m^{-2}}$ per year, or $0.82\,\%$. At those three points, what the one-degree cell fails to see is smaller than the difference between the two products, measured below in figure 15.2.

The grid is declared in the answer. every solar answer carries the field `grid_note`, which declares both grids beside the number. A per-area value presented without it is read as local, and it is not.

The date the series was fetched. the series live in a cache on disk, outside the working directory, keyed on the pair of cells, the period and the parameter list. The cache **does not expire**, so that a figure checked against a stored run stays reproducible. Because POWER reprocesses historical data, a stored series may be a superseded revision. The field `power_provenance` therefore returns, for the daily and the hourly series, whether it was fetched in this run or read from the cache, and at what instant it was obtained.

15.2 What the series measures

The answer carries three irradiance components and two indices. The two indices have similar names and measure different things.

GHI. the total reaching a horizontal surface, summing direct beam and diffuse sky.

DNI. the beam, measured on a plane normal to it. It does not suffer the cosine factor of solar elevation, because the measurement plane follows the Sun.

DHI. the diffuse, on the horizontal plane. It is the part scattered by the atmosphere and by cloud.

Clearness index, K_t. the ratio between GHI and the irradiance that would reach the top of the atmosphere at the same instant. It measures atmospheric transparency, and POWER publishes it.

Clear sky index. the ratio between all-sky irradiation and what the same place would receive under a clear sky, both summed over the window. It measures how much of the available resource the sky delivered.

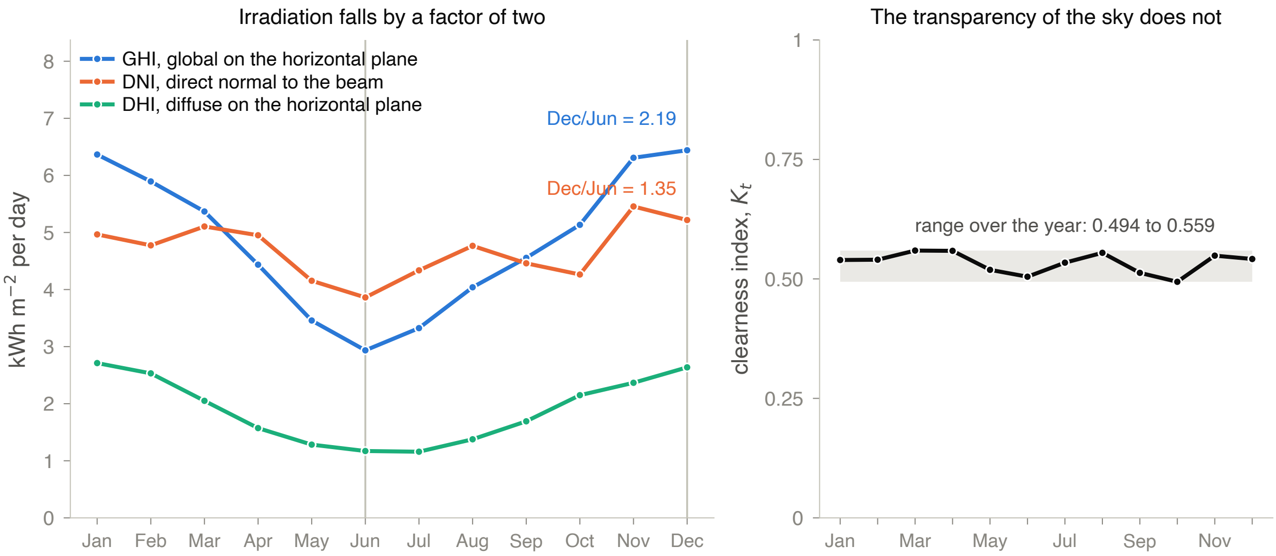

Figure 15.1: The thirty-year climatology in the cell of the three holdings. **Left panel, the three components,** in kilowatt-hours per square metre per day, month by month: GHI in blue, DNI in orange and DHI in green. The two vertical lines mark the month of highest and of lowest GHI, December and June, and the ratio between them is annotated beside the curve it belongs to. **GHI falls by a factor of** 2.19 **between the two months, and DNI falls by** 1.35. **Right panel, the clearness index,** drawn over the interval 0 to 1, which is the range of the index itself. The grey band is the interval between the clearest and the most overcast month of the year. **It runs from** 0.494 **to** 0.559, that is, it barely moves while the irradiation falls by half.

> The annual cycle of the resource is geometric, not atmospheric. If cloud were the cause, the clearness index would fall along with GHI. It does not fall. What is left is the cosine factor: a lower Sun in winter, over the same horizontal surface.
> That settles a design question. Tilting the surface recovers part of the annual cycle, because what the cycle removes is angle. No tilt recovers a loss to cloud.

15.3 Variability between years

Over the daily climatology, the application sums each calendar year and describes the series of totals. The default is thirty years, and POWER publishes up to the last complete year.

In the cell of the three holdings, the mean annual GHI total is $1{,}772\,\mathrm{kWh\,m^{-2}}$, with a standard deviation of 61 and a coefficient of variation of $3.4\,\%$. The tenth percentile is at $1{,}695$ and the ninetieth at $1{,}849$. The lowest of the thirty years was 2015, at $1{,}639$, and the highest was 2020, at $1{,}889$.

A coefficient of variation of $3.4\,\%$ puts the uncertainty of a project at this point in the radiation product and in the system model, not in the variation of the climate between years.
The fitted trend is $+0.83\,\mathrm{kWh\,m^{-2}}$ per year, with $p = 0.531$. **Thirty years of series do not support a claim of increase or of decrease in the resource at this place.**

Only a complete year enters. a year with fewer than 365 daily values is dropped from the series of totals. A partial year sums to less and would enter as a low year, biasing the spread and the trend at the same time.

Fewer than three points receive no line. the slope and the value of p come back as 0 and 1, which reads as an undetectable trend, not as a line fitted to two points.

15.4 Optimal tilt and the tolerance to deviation

Over the hourly series, whose default is ten years, the chain transposes the irradiance onto a tilted plane by the model of Perez et al. (1990). It then sweeps the tilt from $0°$ to $45°$, in steps of half a degree. The azimuth is not swept: it is a field of the request, and the default is $0°$, north.

In the cell of the three holdings, the maximum falls at $25.0°$, and the latitude of the place is $25.1°$. The coincidence with the latitude rule is exact within the step of the sweep, and it is not the useful reading of the sweep.

Table 15.1: The loss from deviating from the optimal tilt, measured over the sweep itself. The column on each side shows that the curve is not symmetric about the maximum: deviating upwards costs slightly less than deviating downwards. The application reports the worse of the two sides at each deviation.

Deviation from the optimum	Loss below the optimum	Loss above the optimum
$5°$	$0.31\,\%$	$0.28\,\%$
$10°$	$1.23\,\%$	$1.15\,\%$
$15°$	$2.74\,\%$	$2.62\,\%$

At the optimum, the plane receives $1{,}883\,\mathrm{kWh\,m^{-2}}$ per year, $8.1\,\%$ above what the horizontal surface receives. Table 15.1 is what the application returns beside that number, and it says the curve is flat.

An installation with $10°$ of tilt error loses $1.2\,\%$ over the year. In the same study, $20°$ of deviation from north costs $0.37\,\%$ and $45°$ costs $2.71\,\%$.
Terrain constraints, roof orientation and structural cost weigh more in the decision than chasing the geometric optimum.

15.5 From the tilted plane to the yield

The conversion into energy uses a reference array of $1\,\mathrm{kW}$ peak, so that the result is a specific yield, independent of the size of the plant. The chain has four steps, and each one comes from a publication.

Incidence angle loss. the direct beam is corrected by the ASHRAE relation. The two diffuse components receive the same relation, integrated over the solid angle each of them occupies, following Marion (2017). The diffuse sky arrives from the whole visible hemisphere, and the diffuse reflected from the ground arrives almost grazing. At $26°$ of tilt, the ground term retains 0.788 against 0.961 for the sky.

Module temperature. by the model of Faiman (2008), with open-rack coefficients. It depends on the irradiance in the plane, the air temperature and the wind.

Direct current power. by the PVWatts model (Dobos, 2014), with a thermal power coefficient of $-0.35\,\%$ per degree, typical of crystalline silicon.

Inversion. by the PVWatts inverter model, with a nominal efficiency of 0.96. All the astronomy, the transposition and the modelling are the reference implementations of `pvlib` (Holmgren et al., 2018).

> The diffuse incidence correction is not an implementation detail. Introducing it at the reference site of the module itself lowered the specific yield by $1.62\,\%$. The size of the effect depends on how diffuse the place is: under a clear sky, with a diffuse fraction of 0.14, it costs $0.75\,\%$. Omitting it is declaring a yield above what the chain supports.

The two performance ratios

The **performance ratio** is the energy delivered in alternating current divided by what the array would deliver at nameplate efficiency. The answer carries two, and swapping them changes the yield by about $7\,\%$.

The modelled ratio. is what this chain produces. Run over the ten-year hourly series of the reference point, it gives 0.858.

The applied ratio. is what multiplies the in-plane irradiation to produce the reported yield. The default is 0.80, and the caller can replace it. The answer declares which of the two was used.

The reported yield is the in-plane irradiation multiplied by the applied ratio, and that calculation is exact by construction, not a correction factor. With the $1{,}883\,\mathrm{kWh\,m^{-2}}$ of the optimum and the reference ratio, the yield comes out at about $1{,}506\,\mathrm{kWh\,kWp^{-1}}$ per year, which corresponds to a capacity factor of $17.2\,\%$. With the modelled ratio, the same plane would give $1{,}616$.

> The study that originated this product reports 0.877 for the same point, and the number the application returns is 0.858. Most of the difference is the diffuse incidence correction described above, which the study chain did not have and which costs the $1.62\,\%$ already cited. The remainder, about half a percentage point, is not explained by it. **The comparison in figure 15.2 is of the study chain**, and the chain the application runs today sits below it, that is, closer to the Atlas.

A constant with a misleading name. the inverter oversizing factor is 1.15, and it is **not** a ratio between direct and alternating current. What `pvlib` receives is the input limit of the inverter, so the array sits at $1{,}000$ watt against $1{,}104$ of inverter, a ratio of 0.906. The code records that correcting the arithmetic to a true ratio of 1.15 would move an already checked yield by $0.09\,\%$, and the label was therefore corrected and the calculation was not.

15.6 Two independent checks

The question that remains is whether to believe the number. It has two halves, and each one was answered by a source that shares no chain with POWER.

The series follows the sky

The first half is whether the series describes the sky of that place. The cloud cover of the Sentinel-2 scenes, which the catalogue already indexes per scene, is an independent observation of the quantity that controls the clear sky index. The test is one of falsification: with no relation between the two, the series does not describe that sky.

Over 134 dates with both a scene and radiation, the rank correlation between cloud and clear sky index is $\rho = -0.850$. In scenes with less than 10 % cloud, the mean clear sky index is 0.952; in those with more than 60 %, it is 0.532. The scale favours the comparison, since cloud cover is reported per 110 km tile and the radiation cell measures about 100 km × 110 km at that latitude.

In fully clouded scenes, the clear sky index varies from 0.1 to 0.9. That is expected. Sentinel-2 samples one instant near 10:30 local time and the index is an integral over the whole day, so an overcast morning followed by a clear afternoon produces exactly that pair. The agreement is monotonic in median, and not predictive date by date.

The absolute values, against the Global Solar Atlas

The second half is whether the values are right. A series can follow the cloudiness and still be biased, and answering that requires an external reference. The Global Solar Atlas (ESMAP, 2020) serves: a distinct geostationary retrieval chain, a resolution of 250 m for irradiance and 1 km for generation, publishing the same quantities.

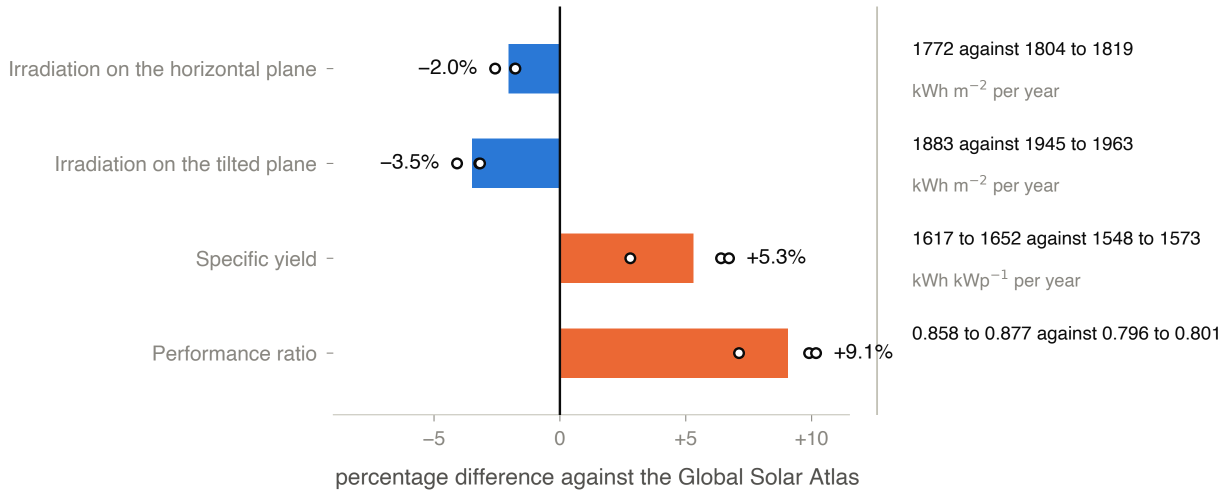

Figure 15.2: The difference between the study chain and the Global Solar Atlas, in the four quantities both products publish. **Each bar is the mean over the three holdings** and the white points on it are the three, one by one. To the right of each line are the absolute values on both sides, in the order *study against Atlas*. The chain the application runs today is more recent than that of the study and returns a performance ratio of 0.858, below the 0.877 drawn here and therefore closer to the Atlas. **The two resource quantities sit below the reference,** in blue: the irradiation on the horizontal plane by 2.0 % and on the tilted plane by 3.5 %. **The two generation quantities sit above it,** in orange: the specific yield by 5.3 % and the performance ratio by 9.1 %. The optimal tilt is 25° in both products, and therefore does not appear as a bar.

The characterisation of the resource checks out. GHI falls within 2.6 % between two products that share neither sensor nor model, and the optimal tilt is the same.
The generation estimate is high, and the excess is entirely in the performance ratio: 0.877 from the study against 0.798 from the Atlas. That is what is expected of a chain that models neither soiling, nor inter-row shading, nor degradation, nor availability, nor wiring. The limitation stops being a statement and acquires a size, about 6 % in that configuration.

It is that measurement which fixes the reference ratio of 0.80 as the application default. It sits close to the Atlas value, which includes the loss terms this chain omits.

15.7 The module temperature depends on the reanalysis wind

One of the three holdings came out with a yield $2.1\,\%$ below the other two. Since the irradiance is identical in all three, the difference could only come from the module temperature. It did not come from the air temperature: the daytime difference between the cells is $0.16\,°\mathrm{C}$, against $0.81\,\mathrm{m\,s^{-1}}$ of wind difference.

> The MERRA-2 wind field in the cell of that holding has $97.3\,\%$ of its hours below $0.5\,\mathrm{m\,s^{-1}}$ and a maximum of $1.08\,\mathrm{m\,s^{-1}}$ over ten years. Replacing the wind with a value common to all three, the yield spread between them falls from $2.12\,\%$ to $0.02\,\%$.
> **The difference was not in the place. It was in the wind field.**

The bias is not confined to that cell. The wind of the other two has a mean of $0.75\,\mathrm{m\,s^{-1}}$ and a maximum of $2.94\,\mathrm{m\,s^{-1}}$ over ten years, low values for open farmland. All three estimates carry that bias, and one of them merely makes it visible.

> Two areas falling in different meteorological cells can return different yields without the places differing in any way. The chain has no way of separating the two causes, because no surface wind measurement enters it.

15.8 What cannot be concluded

That the number describes the drawn area. it describes the one-degree cell its centroid fell in. Two areas inside the same cell return identical radiation series, measured over $10{,}958$ days.

That the reported yield is a plant forecast. the chain omits soiling, inter-row shading, degradation, availability and wiring. Against the Global Solar Atlas, the estimate sits $2.8\,\%$ to $6.7\,\%$ above.

That the two performance ratios are the same thing. the modelled one is what the chain produces and the applied one is what multiplies the in-plane irradiation. Between 0.858 and 0.80 there is $7.3\,\%$ of yield.

That the absolute value was checked against a surface measurement. there is no pyranometer at any of the sites. The check with Sentinel-2 shows that the series follows the variation of the sky, which is not the same as checking its value.

That tracking has been evaluated. the sweep is for a fixed system. A tracking axis would change the result appreciably at this DNI level, and it was not measured.

That comparing the yield of two areas compares the places. the difference may come from the reanalysis wind field, as the previous section measures.

CHAPTER 16

The energy model and the loss waterfall

Chapter 15 ended with two performance ratios and a difference between them that was not explained, only named. This chapter opens it.

The energy chain fetches no new data. It consumes the hourly series, the transposition and the photovoltaic model the previous chapter described. And it does something no other product in the application does: it **declares, step by step, what it measured and what it assumed**.

16.1 Three performance ratios, and one external check

The answer carries three ratios. They are not versions of one number.

The modelled one. is what this chain computes: the product of three factors measured over the hourly series.

The derived one. is the modelled one multiplied by the PVWatts loss terms the chain does **not** model, each with a declared value.

The applied one. is what multiplies the in-plane irradiation to produce the reported yield. The default is 0.80, the value checked against the Global Solar Atlas in chapter 15.

> The derived ratio **does not replace** the applied one, and the reason is declared in the code: the derived ratio has no external check. It is the decomposition of this chain plus the assumptions it declares, and those two things are not the same.

One resolves, all consume. there is a single point at which a performance ratio is decided, and every other function receives the record it returns, not a loose number. A loose number carries neither provenance nor reporting basis, and two products fed different literals would disagree on the same screen with nothing to explain it. The boundary of each product refuses the loose number.

16.2 The waterfall: what is measured and what is declared

The base of the waterfall is the global horizontal irradiation **of the hourly window**, not the thirty-year climatology. The two appear side by side in the figure because they differ: $1{,}783\,\mathrm{kWh\,m^{-2}}$ against $1{,}772\,\mathrm{kWh\,m^{-2}}$, $0.65\,\%$ of difference.

> Starting the waterfall from the climatology would leave **every** ratio below it wrong by the difference between the two windows. The application labels the two instead of reconciling them, because they are different products over different periods.

The step that is not a plant loss

The second step of the waterfall costs $2.33\,\%$ and is not a loss at all.

NASA POWER publishes GHI, DNI and DHI, and the three **do not close**: the horizontal plane reconstructed from the direct and the diffuse does not reproduce the published global. The chain uses the reconstruction, because that is what the transposition starts from, and carries the difference as a line of its own, outside the performance ratio.

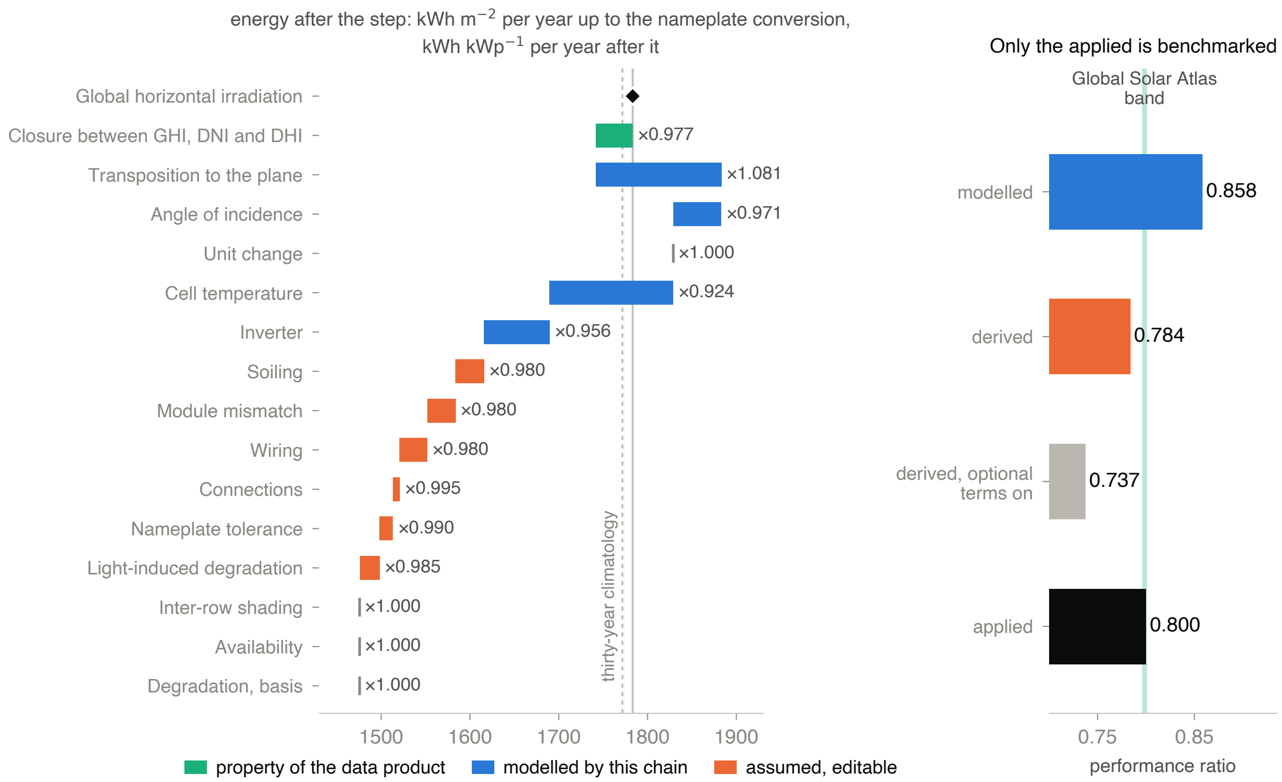

Figure 16.1: A run of the energy chain over ten years of hourly series, at the reference point of chapter 15. **Left panel, the waterfall:** each bar runs from the energy before the step to the energy after it, and the factor of the step is annotated beside it. **The colour states the nature of the step**, and it is the reading of the figure: blue, measured by this chain over the series; orange, a declared value it does not model and the user can edit; green, a property of the radiation product, which sits outside the performance ratio. The diamond marks the base, and the dashed line to its left is the thirty-year climatology, which is another window and another product. The last three lines have no bar because they are at their default, which is a factor of 1.000. The unit changes on the line *Unit change*, and the two halves are numerically continuous only because the reference array is 1,000 W at 1,000 W m^{-2}. **Right panel, the four performance ratios** the answer carries, against the grey-green band the Global Solar Atlas implies over the same sites. The modelled one is 0.858, the derived one 0.784, the derived one with both optional terms switched on at the values PVWatts suggests is 0.737, and the applied one is 0.800.

It is not a low-Sun artefact. Restricting to solar elevations above $10°$, the residual stays of the same order. It is a property of the radiation product, and a performance ratio that absorbed it would be charging the plant for a residual of the data.

The three measured factors, and the identity that is enforced

Between the in-plane irradiation and the alternating current energy there are three factors this chain measures: incidence ($0.971{,}3$), cell temperature ($0.923{,}9$) and inverter ($0.956{,}3$).

Their product is the modelled ratio, $0.858{,}2$, by construction. The application does not trust that: it computes the residual between the product of the three and the direct ratio, and **raises an error** if it exceeds 1×10^{-9}. In the run in the figure the residual is of the order of 1×10^{-16}.

Why raise instead of warn. a residual different from zero means the frame did not come from a single chain. If that happens, every number below it is wrong, and a warning would be read as a caveat.

The six declared terms

After the measured ones come six values the chain does not model and does declare: soiling $2\,\%$, module mismatch $2\,\%$, wiring $2\,\%$, connections $0.5\,\%$, nameplate tolerance $1\,\%$ and light-induced degradation $1.5\,\%$. Together they are worth $8.68\,\%$, and all of them are editable.

Two further terms exist and are **zero by default**: inter-row shading and plant availability.

The zero default is not an omission, and the justification is measured. Switching them on at the $3\,\%$ PVWatts suggests takes the derived ratio to 0.737, which is $7.4\,\%$ *below* the floor of the Atlas band. There is no row geometry in this chain, so there is nothing from which to compute a shading between rows.

The losses are shown multiplied into the total. and not at the physical position of each. The cost of that presentation was measured: placing them physically gives $1{,}476.20\,\mathrm{kWh\,kWp^{-1}}$ against $1{,}474.73\,\mathrm{kWh\,kWp^{-1}}$ in the flat form, $0.10\,\%$. The energy of each line is a presentation of the stack of factors, not a simulation run again.

16.3 The largest measured loss, and the wind at the wrong height

Cell temperature is the largest modelled loss in the waterfall: $0.923{,}9$, or $7.6\,\%$. It depends on the in-plane irradiance, the air temperature and the wind, and that is where the problem is.

The wind coefficient of the model of Faiman (2008) is calibrated against wind measured **at module height**. This chain feeds the model the $2\,\mathrm{m}$ wind from the reanalysis.
It is the wrong height for that coefficient, and the bias is systematic, **independent of any defect of the site**. It sits inside the modelled ratio, which is where every energy figure in this chapter is anchored.

This is distinct from what chapter 15 measured. There, the wind field of one cell was implausible and produced a difference between sites that did not exist. Here, the measurement height is wrong at every site, including those where the wind field is plausible. The application declares the bias and does not quantify it, because it has nothing to measure it against.

The module type is also a choice, not a measurement. the thermal power coefficient comes from a selection among three PVWatts types: $-0.004{,}7$, $-0.003{,}5$ and $-0.002{,}0$ per degree. The application uses the second, which is the **least conservative** option among the crystalline silicon ones, and it therefore names the selection in every answer and carries the alternatives beside it.

16.4 Degradation is a reporting basis, not a loss line

The chain offers two bases. On the *year one* basis the factor is 1.000. On the *lifetime average* basis it is the mean of $(1-r)^t$ over the analysis period, which at the default of $0.5\,\%$ per year and twenty-five years gives $0.942{,}2$, or $5.78\,\%$ less.

> The two bases are comparable neither with each other nor with an external reference computed on the other. A lifetime average yield read against a year one exceedance band mixes two conventions, and every energy figure therefore prints the basis it was computed on.

Zero per year is a value. a request declaring zero degradation is a statement, not an omission. The code records that, when this was treated as an absence, the default of $0.5\,\%$ came back underneath and multiplied every lifetime average figure by $0.942{,}2$ instead of 1.0.

16.5 Tracking: the answer inverts with the basis

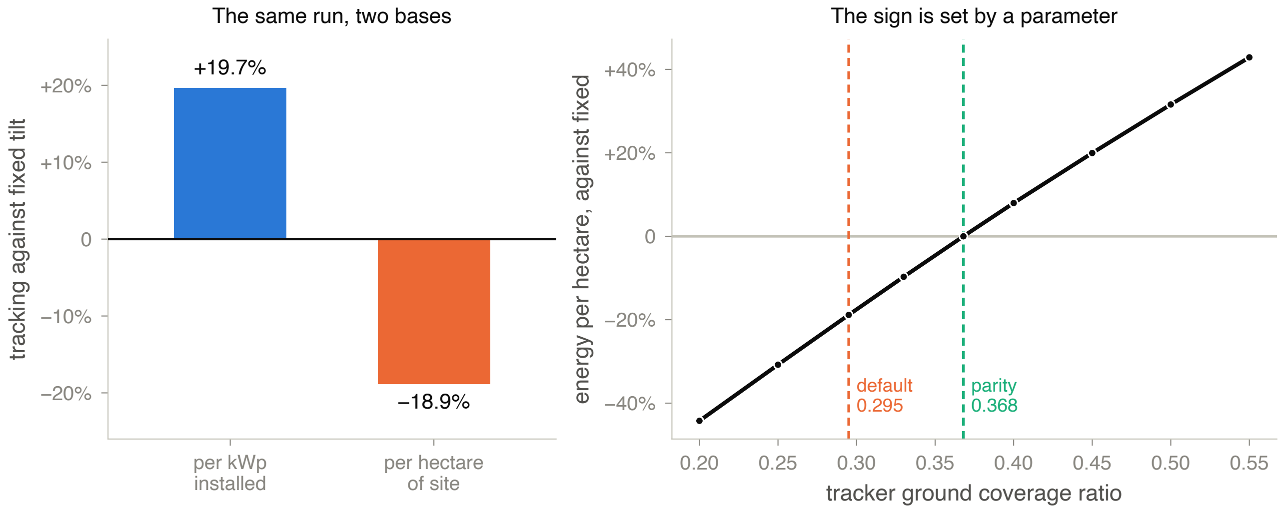

Figure 16.2: Single-axis horizontal tracking against fixed tilt, in the same run as figure 16.1. **Left panel:** the same series, the same performance ratio and the same site, on two bases. Per kilowatt-peak installed the tracker delivers $19.7\,\%$ more; per hectare of site, with the pair of ground coverage ratios the application carries by default, it delivers $18.9\,\%$ less. **Right panel:** the energy per hectare of the tracker relative to that of the fixed array, against the ground coverage ratio of the tracker. The grey line is parity. The dashed orange line is the application default, 0.295; the green one is the ratio at which the curve crosses parity, 0.368, which the product resolves numerically. **The sign of the per-hectare comparison is fixed by that parameter**, and not by the site.

> Per kilowatt-peak the tracker wins and does not invert in any configuration tested. Per hectare it loses, with the default pair of coverage ratios.
> The two bases answer different questions. Whoever reads only the first thinks they have read the answer.

The published measurements disagree on the sign

The per-hectare question has been measured directly over built plants, by two sources, and they do not agree.

Bolinger and Bolinger (2022) publishes energy density over the direct area of the array: $447\,\mathrm{MWh}$ against $394\,\mathrm{MWh}$ per acre per year, or $-11.9\,\%$ for tracking.

Ong et al. (2013) publishes the variation in land use intensity. At the sites of direct irradiance closest to this one, it runs from $-4.9\,\%$ to $-2.7\,\%$, and the negative sign means that tracking uses *less* land per unit of energy.

> The two measure the same question and point in opposite directions. The application reports them separately, each at the precision of its source, and **does not average them**. The line derived from the model is a third one, presented with the pair of parameters that produced it.

Where the coverage ratios come from. they are not numbers without a source. They come from the capacity densities published by Bolinger and Bolinger (2022), converted to hectares and divided by the nameplate power of one hectare fully covered at the module efficiency adopted. The result is $0.432{,}4$ for the fixed array and $0.296{,}5$ for the tracker. The application defaults are those values to three places, and the answer reports the residual. The tracker default used to be 0.35, with no source, $18\,\%$ above what that same work implies, which moved the per-hectare headline by about fourteen percentage points.

The annual gain is a summer gain. in the run in the figure, the $19.7\,\%$ annual decomposes into $36.1\,\%$ in summer and $1.5\,\%$ in winter. Anyone sizing for a winter-limited load reads the annual number for a season in which the comparison is nearly neutral.

What is not in the model. motor consumption, controller availability and mechanical stoppage, all of which the tracker carries and the fixed array does not. Capital and operating cost are not there either. The comparison is one of energy, and it does not support a design decision on its own.

16.6 When the energy arrives

The generation profile answers the distribution across the day and the year, and three details of it decide whether it can be read.

The hour is UTC unless requested otherwise. POWER values are hourly means labelled by the starting hour, in UTC. A shift to local time is applied only if the caller states it, and the answer declares which was used. Without that, a midday peak falls at the wrong hour.

The full-sun hours are in the plane of the module. and not in the horizontal plane, which is the plane most sources publish them on. It is the same plane the yield is computed in.

The level is not a yield. the profile is computed over the modelled alternating current series **before** any performance ratio. The shape is the shape of the model; the level is not a reportable number.

16.7 From area to capacity, and the basis that decides

The suitable area enters the chain from outside, and its conversion into installed capacity passes through a density that has to be declared alongside.

Seven bases, and they do not measure the same thing. between the lowest and the highest there are $0.427{,}3$ and $0.864{,}9\,\mathrm{MW}$ of direct current per hectare, **a factor of two**. The difference lies in two choices: direct array area against total site area, and fixed against tracker.

> A suitability raster classifies whole terrain, including what would become road, spacing and substation. Its hectares are *total site* area. Applying a *direct array area* density to them treats the whole polygon as module footprint.

The buildable fraction is an example, not a median. the value of 0.75 that converts direct area into total area is the worked example of the article itself, not a median measured over the fleet.

Two similar ratios that are not the same. the fleet direct-to-alternating current ratio, $1.314{,}1$, serves **only** to convert a density published in alternating current into direct current. It is not the inverter oversizing factor chapter 15 described. They are unrelated quantities, and the numerical proximity is a coincidence.

The three classes are never summed. the suitable area with no use conflict is the only one that feeds the headline. The suitable area in conflict with annual cropland is reported separately, because it is not additional capacity: it is the same land, counted against food production. The area of restrictive slope is reported only as area, because the density references do not cover the structure it would require.

A scattered hectare does not host the plant. beside every capacity figure comes the largest contiguous block of the class and how many blocks exist, with eight-way connectivity. An area summed from many small patches does not support the arithmetic that converted it into megawatts.

Shading enters on the wrong basis unless converted. the horizon loss the terrain product publishes is a fraction of the **direct** irradiation. A total in the plane of the module is direct plus diffuse, so the loss has to be scaled by the direct fraction before it reaches the yield. At the reference site, with a direct fraction of $0.642{,}5$, a mean loss of $3\,\%$ in the beam is worth $0.980{,}72$ over the total, not $0.970{,}00$. Without the direct fraction declared, the application **refuses** to apply the derate rather than applying it on the wrong basis.

16.8 The exceedance band, and what it does not contain

Over the annual totals, the chain returns exceedance factors. In the run in the figure, with thirty complete years, P90 is $0.956{,}5$ of the mean: the whole band, from P50 to P90, is worth $4.35\,\%$.

> **P90 is the value exceeded in ninety per cent of years, and it therefore sits *below* P50.** The resource card reports statistical percentiles, in which the ninetieth is the high value. The two conventions appear on the same screen, and the answer carries the correspondence table: the P90 exceedance and the tenth statistical percentile are the same estimate of the same quantity.

The estimator applied is the empirical one, the same the resource card uses, so that the two cannot contradict each other. The normal fit is reported alongside, with a Shapiro-Wilk test of the normality it assumes, and not applied. In this run the test gives $p = 0.916$.

> The band propagates **one** source of uncertainty: the measured variability of the resource between years.
> Eight terms stay outside, and the answer names them one by one. The bias of the radiation product against surface measurement, the error of the transposition and photovoltaic model, the uncertainty of the performance ratio, the variation in soiling, degradation, availability, inter-row shading and the capacity density basis.
> It is therefore **narrower than a project finance P90**, and the answer states this in the field itself.

And there is a comparison that decides how to read the band. The choice of density basis moves the result by $102\,\%$; the whole exceedance band moves it by $4.35\,\%$.

16.9 What cannot be concluded

That the derived ratio is better than the applied one. the derived one is the decomposition of this chain plus the declared assumptions, and it sits $1.55\,\%$ below the floor of the Atlas band. The applied one is the only one with an external check. Both are correct under their own assumptions.

That the modelled ratio can be quoted as a plant performance ratio. it includes none of the six declared terms, and it is $7.3\,\%$ above the applied one. It is the yield of a plant with no soiling, no module mismatch, no wiring loss and no nameplate tolerance.

That the tracker gain is what the headline says. it depends on the basis. In the same run it is $19.7\,\%$ per kilowatt-peak and $-18.9\,\%$ per hectare, and the two published measurements of the per-hectare question disagree with each other on the sign.

That the reported capacity is interconnectable. it is a ceiling of resource and of land. There is no grid consultation, no licensing and no check of legal reserve, and the class areas inherit the suitability conventions they were produced under.

That the exceedance band is the uncertainty of the project. it contains only the variability of the resource between years. Eight declared terms stay outside, and one of them, the density basis, moves the result twenty-three times more than the whole band.

That the energy is linear in the annual irradiation. the exceedance treats it that way, and that is first order. The ratio between the tilted and the horizontal plane depends on the diffuse fraction, which varies between years, and the photovoltaic output is slightly sublinear in irradiance through a temperature effect.

References

Barnes, R., Lehman, C., Mulla, D. (2014). Priority-flood: an optimal depression-filling and watershed-labeling algorithm for digital elevation models. *Computers & Geosciences*, 62, 117–127.

Bolinger, M., Bolinger, G. (2022). Land requirements for utility-scale PV: an empirical update on power and energy density. *IEEE Journal of Photovoltaics*, 12(2), 589–594. A sample of 736 plants and 35,482 MW of direct current, in commercial operation between 2007 and 2019, in the United States.

Brown, L. D., Cai, T. T., DasGupta, A. (2001). Interval estimation for a binomial proportion. *Statistical Science*, 16(2), 101–133.

Dobos, A. P. (2014). *PVWatts version 5 manual*. NREL/TP-6A20-62641, National Renewable Energy Laboratory, Golden, CO.

Doelling, D. R., Haney, C. O., Scarino, B. R., Gopalan, A., Bhatt, R. (2016). Advances in geostationary-derived longwave fluxes for the CERES synoptic (SYN1deg) product. *Journal of Atmospheric and Oceanic Technology*, 33(3), 503–521.

European Space Agency (2021). *Sentinel-2 Products Specification Document*, PSD v14.9. Processing baseline 04.00, in production since 25 January 2022.

Energy Sector Management Assistance Program (2020). *Global Solar Atlas 2.0*. World Bank, Washington, DC. Data modelled by Solargis.

Faiman, D. (2008). Assessing the outdoor operating temperature of photovoltaic modules. *Progress in Photovoltaics: Research and Applications*, 16(4), 307–315.

Feyisa, G. L., Meilby, H., Fensholt, R., Proud, S. R. (2014). Automated Water Extraction Index: a new technique for surface water mapping using Landsat imagery. *Remote Sensing of Environment*, 140, 23–35.

Gelaro, R., McCarty, W., Suárez, M. J., et al. (2017). The Modern-Era Retrospective analysis for Research and Applications, version 2 (MERRA-2). *Journal of Climate*, 30(14), 5419–5454.

Gretton, A., Borgwardt, K. M., Rasch, M. J., Schölkopf, B., Smola, A. (2012). A kernel two-sample test. *Journal of Machine Learning Research*, 13, 723–773.

Holmgren, W. F., Hansen, C. W., Mikofski, M. A. (2018). pvlib python: a python package for modeling solar energy systems. *Journal of Open Source Software*, 3(29), 884.

Jordan, D. C., Kurtz, S. R. (2013). Photovoltaic degradation rates: an analytical review. *Progress in Photovoltaics: Research and Applications*, 21(1), 12–29. The rate the application adopts is the median for crystalline silicon.

Kopp, G., Lean, J. L. (2011). A new, lower value of total solar irradiance: evidence and climate significance. *Geophysical Research Letters*, 38, L01706.

Marion, B. (2017). Numerical method for angle-of-incidence correction factors for diffuse radiation incident photovoltaic modules. *Solar Energy*, 147, 344–348.

McFeeters, S. K. (1996). The use of the Normalized Difference Water Index (NDWI) in the delineation of open water features. *International Journal of Remote Sensing*, 17(7), 1425–1432.

Melo, J. L. S., Magalhães, D. K., Kolodziej, J. E., Kuhn, E. V. (2026). Automatic Land Cover Classification with Sentinel-2 and MapBiomas Time Series. *XLIV Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT)*, Salvador, BA, Brazil.

Nobre, A. D., Cuartas, L. A., Hodnett, M., Rennó, C. D., Rodrigues, G., Silveira, A., Waterloo, M., Saleska, S. (2011). Height Above the Nearest Drainage: a hydrologically relevant new terrain model. *Journal of Hydrology*, 404(1–2), 13–29.

Olofsson, P., Foody, G. M., Herold, M., Stehman, S. V., Woodcock, C. E., Wulder, M. A. (2014). Good practices for estimating area and assessing accuracy of land change. *Remote Sensing of Environment*, 148, 42–57.

Ong, S., Campbell, C., Denholm, P., Margolis, R., Heath, G. (2013). *Land-use requirements for solar power plants in the United States*. NREL/TP-6A20-56290, National Renewable Energy Laboratory, Golden, CO.

Otsu, N. (1979). A threshold selection method from gray-level histograms. *IEEE Transactions on Systems, Man, and Cybernetics*, 9(1), 62–66.

Perez, R., Ineichen, P., Seals, R., Michalsky, J., Stewart, R. (1990). Modeling daylight availability and irradiance components from direct and global irradiance. *Solar Energy*, 44(5), 271–289.

Pontius, R. G., Millones, M. (2011). Death to Kappa: birth of quantity disagreement and allocation disagreement for accuracy assessment. *International Journal of Remote Sensing*, 32(15), 4407–4429.

Ramdas, A., Reddi, S. J., Póczos, B., Singh, A., Wasserman, L. (2015). On the decreasing power of kernel and distance based nonparametric hypothesis tests in high dimensions. *AAAI Conference on Artificial Intelligence*, 29(1).

Strobl, C., Boulesteix, A.-L., Zeileis, A., Hothorn, T. (2007). Bias in random forest variable importance measures: illustrations, sources and a solution. *BMC Bioinformatics*, 8, 25.

Xu, H. (2006). Modification of Normalised Difference Water Index (NDWI) to enhance open water features in remotely sensed imagery. *International Journal of Remote Sensing*, 27(14), 3025–3033.

APPENDIX A

The sidecar actions

Everything TERRA computes passes through a separate Python process, the *sidecar*. It is not a service. The Go side writes **one** JSON object on standard input and reads **one** JSON result on standard output; progress and errors come out as JSON lines on standard error. One field of the request, `action`, chooses what will be answered.

The list is the exact outline of what the application can do. Chapter 1 states that there is no map algebra in TERRA; the table below is where that statement is checked, in one catalogue.

Table A.1: The seventeen actions, grouped by product. The order is that of the `ACTIONS` table in the code, regrouped for reading.

`action`	What it answers
Land cover	
`predict`	the complete classification: scenes, features, model, map, per-class statistics, index series and phenology. When the area intersects Brazil, also the comparison with MapBiomas
`lulc`	only the descriptive MapBiomas analysis, without classifying
`list_datacube`	which scenes the window and the filters return, without reading a single band
`render_composite`	a band composite to draw on the map
Error and transfer	
`domain_shift`	the distance between two runs, from the fingerprints stored at classification time
`domain_shift_cohort`	the same measure over a set of runs, rather than over a pair
Water and terrain	
`water`	the water mask by spectral index, date by date
`flood_envelope`	the flood envelope: the HAND chain over four elevation products, and the agreement raster between them
Energy	
`solar_resource`	the solar resource at the centroid of the area
`solar_terrain`	the irradiance distributed over the slope and aspect of the terrain
`solar_siting`	where inside the area a plant can be placed
`energy_model`	the yield, with every loss term declared
`wind_resource`	the wind resource screening, with the diagnostics that qualify it
Canopy	
`canopy_field`	the simulated stand: age, leaf area and the light balance across the season
`canopy_from_aoi`	the same, starting from the series the area has already produced
`canopy_mesh`	the three-dimensional canopy mesh, for drawing
Service	
`ping`	whether the interpreter imports what the sidecar needs

> A request that names no action runs the classification. A request that names a **nonexistent** action does the same, because the dispatch is `ACTIONS.get(action, action_predict)`. It is the behaviour inherited from the chain of `if` statements the table replaced, and the code documents it as deliberate.
> For anyone calling the sidecar from a script, the consequence is that a typing error in `action` does not return an error. It returns a classification, which costs minutes and is not what was asked for.

What is not here. the request fields of each action. They stay in the product chapter and, in canonical form, in `internal/analysis/types.go` of *geosense-infer*. A field table here would age with every new field, and a wrong reference is worse than an absent one.

Known divergence. the `docs/API.md` of *geosense-infer* lists four of these actions. The table above is the one in the code.

APPENDIX B

Provenance, chapter by chapter

Every statement in this book comes from a file, a study or a publication. This table is the complete list, chapter by chapter.

It answers a question the body of the text does not answer well: **how much of this book can you open on your own?** The third column says where each entry lives, and the count below is the summary of it.

Table B.1: Where the 119 provenance entries of this book live.

Where it lives	Entries	What that means for the reader
Application	85	*geosense-infer*, public under GPL v3. A file, function or constant that opens without asking anyone for anything
Book	15	The figure scripts of this book and the data they cache
Line	10	*TERRA-Research*, private. Named by nickname, and the study can be requested; section 1.1 says from whom
Published	9	Literature, in the bibliography of this book

Of 119 entries, 85 point at the public code of the application and 10 at the private line. What supports this book is, almost everywhere, something the reader can open.

Table B.2 is printed on a landscape page because the middle column carries file paths and function names, which do not hyphenate.

Table B.2: The provenance of each chapter. *What* is the element; *where it lives* is the column of table B.1; *where it is* is the file, the function or the study.

Ch.	What	Where it lives	Where it is
Chapter 1 *What TERRA delivers, and where it comes from*			
1	Product scope	Application	`README.md` of *geosense-infer*, version 0.4.0
1	The 17 actions	Application	the `ACTIONS` table in `sidecar/infer.py`
1	Wind qualification	Application	`RESULT_QUALIFIER` and `data_quality` in `sidecar/wind.py`
1	Solar check	Application	the note on the Global Solar Atlas in the opening of `sidecar/solar.py`
1	Tables 1.1 and 1.2	Line	study index and triage notes of the *TERRA-Research* line, August 2026
Chapter 2 *The common protocol: area, catalogue and window*			
2	Catalogue and bands	Application	`list_stac_products` in `sidecar/infer.py`
2	Monthly cadence	Application	the `monthly_best` block of the same function
2	Resampling	Application	`load_band_to_reference_grid`, bilinear
2	The 22 dates	Application	`build_feature_matrix`, and the `n_dates_model` of the prediction action
2	The middle scene	Application	`products[len(products) // 2]`, in the *Temporal Transformer* and Prithvi paths
2	Elevation, flood	Application	`COLLECTIONS` and `DEFAULT_IDS` in `sidecar/dem.py`
2	Elevation, solar	Application	`DEM_COLLECTION` and `fetch_dem` in `sidecar/solar.py`
2	Interface defaults	Application	`maxCloud` and `monthlyBest` in `frontend/src/App.tsx`
Chapter 3 *From the stored number to the number the model sees*			
3	The constants	Application	in `sidecar/infer.py`: `QUANTIFICATION_VALUE`, `BOA_ADD_OFFSET`, `BOA_OFFSET_FIRST_BASELINE`, `BOA_OFFSET_FIRST_DATE`
3	The two conversions	Application	`to_reflectance` and `as_trained`, in the same file, with the docstrings that justify the seam
3	Training set	Application	22 products, 2024-05-04 to 2026-01-04, tiles T21JZN and T22JBT, declared in the docstring of `as_trained`
3	The 2021 vs 2024 measurement	Line	correction note of the *TERRA-Research* line on the reflectance offset
3	Reference	Published	ESA (2021), for the baseline
Chapter 4 *The three model paths, and the fixed legend*			
4	The five classes	Application	`model/label_encoder.joblib`, vector `[3, 21, 25, 39, 41]`

Table B.2, continued

Ch.	What	Where it lives	Where it is
4	The 80 features	Application	`model/feature_names.joblib`
4	How they are assembled	Application	`build_feature_matrix` in `sidecar/infer.py`
4	The application default	Application	`model_kind` falling back to `spectral`, in `sidecar/infer.py`
4	The Transformer	Application	`NUM_FRAMES`, `N_BANDS` and `band_specs` in `sidecar/temporal_transformer.py`
4	The mode rule	Application	`modeBlockedBy` in `frontend/src/lib/classifyOptions.ts`
4	The two figures	Book	`figuras/figuras.py`, which reads the artefact and the ablation CSV without transcribing a number
4	The ablation	Line	representation study of the *TERRA-Research* line, scenarios A and pooled, over real MapBiomas
Chapter 5 *What the map claims, and in which unit*			
5	The per-class statistics	Application	`class_statistics` in `sidecar/infer.py`
5	The cell size	Application	`reference_pixel_size_m`, in the same file, read from the grid transform
5	The reference grid	Application	built from B04 at its native resolution, in the prediction path
5	Confidence, RF	Application	`classify_from_features` in `sidecar/infer.py`
5	Confidence, TT	Application	`predict_pixels` in `sidecar/temporal_transformer.py`
5	The floor of the scale	Application	`confidence_floor`, with the comment justifying reporting it beside the mean
5	The temporal mode	Application	the cumulative stack loop of the prediction action
5	Figure 5.1	Book	`figuras/figuras.py`, which reads the number of classes from the embedded label encoder
Chapter 6 *Agreement with MapBiomas: the matrix and its intervals*			
6	The comparison	Application	`agreement_against_reference` in `sidecar/lulc.py`
6	The sampling unit	Application	the docstring of the same function, which records the factor of nine pixels per cell
6	The three accuracies	Application	the per-class loop of the same function, with `producers_pct` and `users_pct`
6	The interval	Application	`_wilson_interval`, in the same file
6	Outside the legend	Application	the `outside` counter, reported as `n_outside_legend`
6	Figure 6.1	Book	`figuras/figuras.py`, which runs the interval function of the sidecar itself instead of reimplementing it
6	Reference	Published	Brown et al. (2001), on why Wilson and not Wald

Table B.2, continued

Ch.	What	Where it lives	Where it is
Chapter 7 *Quantity and allocation: two errors that do not add up*			
7	The decomposition	Application	the quantity and allocation block in `agreement_against_reference`, in `sidecar/lulc.py`
7	The payload fields	Application	`quantity_disagreement_pct` and `allocation_disagreement_pct`
7	Why kappa stays out	Application	the comment immediately above the block, in the same file
7	The defect it corrects	Application	the docstring of the same function, which records the composition panel showing the quantity term on its own
7	Figure 7.1	Book	`figuras/figuras.py`, which extracts the decomposition block from the sidecar and runs it over synthetic matrices
7	Reference	Published	Pontius and Millones (2011)
Chapter 8 *Agreement by block: where the disagreement is*			
8	The grid and the floor	Application	`BLOCKS_PER_SIDE` and `MIN_BLOCK_CELLS` in `sidecar/lulc.py`, each with its justification in the comment
8	The statistics	Application	`_block_agreement`, in the same file
8	Not being a test	Application	the docstring of that function, which declares the spread as descriptive
8	Figure 8.1	Book	`figuras/figuras.py`, which reads the number of blocks per side and the floor from the sidecar itself
8	Reference	Published	Olofsson et al. (2014), for what an assessment with a sampling design would be
Chapter 9 *Class separability: contrast and signature*			
9	The two distances	Application	`bhattacharyyaGaussian` and `jeffriesMatusita`
9	Where they live	Application	`frontend/src/lib/separability.ts`
9	The two readings	Application	the opening of the same file, which separates contrast from signature
9	The maximum over bands	Application	the `best` field of the same implementation, with the reason in the comment
9	The per-class spectrum	Application	`class_spectra` in `sidecar/infer.py`, with `SPECTRUM_MIN_PIXELS`
9	The seven bands	Application	`TERRA_BANDS`, in the same file
9	Figure 9.1	Book	`figuras/figuras.py`, which runs the functions of the frontend itself instead of reimplementing them
9	The pair 0.059 and macro-F1 0.81	Line	multi-source ceiling study of the *TERRA-Research* line

Table B.2, continued

Ch.	What	Where it lives	Where it is
Chapter 10 *Domain shift: the diagnosis*			
10	The fingerprint	Application	`build_fingerprint` in `sidecar/domain_shift.py`
10	The three distances	Application	`kl_divergence`, `cva_magnitude` and `mmd_rbf`, in the same file
10	Why the kernel is Gaussian	Application	the docstring of `mmd_rbf`, which records the redundancy and the blindness of the linear one
10	The per-feature table	Application	`feature_shift_table`
10	The standardisation	Application	`_standardise`, in training deviations
10	Figure 10.1	Book	`figuras/figuras.py`, which imports the sidecar module and calls both functions
10	References	Published	Gretton et al. (2012) for the characteristic kernel, Ramdas et al. (2015) for the loss of power with dimension, Strobl et al. (2007) for the bias of impurity importance
Chapter 11 *Domain shift: what it does not correct*			
11	The six families	Line	real transfer study of the *TERRA-Research* line, Iowa and western Paraná pair
11	The constructed experiment	Line	synthetic study of the same line, with the shift assembled by named mechanism
11	The ceiling with declaration labels	Line	multi-source architecture study, which swaps the target reference
11	Adaptation out of scope	Application	the opening of `sidecar/domain_shift.py`
11	Figure 11.1	Book	`figuras/figuras.py`, which reads the CSV of the real study and the table from the report of the synthetic one
Chapter 12 *Surface water by spectral index*			
12	The three indices	Application	`compute_water_indices` in `sidecar/water.py`
12	The threshold	Application	`otsu_threshold` and `otsu_threshold_for_date`, in the same file
12	Validity per date	Application	`per_date_valid_mask`, whose docstring records the measurement in figure 12.1
12	The occurrence	Application	`occurrence_map` and `classify_occurrence`
12	References	Published	McFeeters (1996) for NDWI, Xu (2006) for MNDWI, Feyisa et al. (2014) for AWEI and Otsu (1979) for the threshold
Chapter 13 *The terrain chain: from the grid to HAND*			
13	The chain	Application	`sidecar/hand.py`, with the five analytical checks in `sidecar/tests/test_hand.py`

Table B.2, continued

Ch.	What	Where it lives	Where it is
13	The filling	Application	`fill_depressions`, Priority-Flood with an epsilon
13	The flow direction	Application	`d8_receivers`, and the positive infinity frame
13	The height	Application	the `hand` function, which measures against the original DEM
13	The four products	Application	`COLLECTIONS` and `DEFAULT_IDS` in `sidecar/dem.py`
13	The margin and the merge	Application	the opening of `sidecar/dem.py`
13	Figure 13.1	Book	`figuras/figuras.py`, which runs the sidecar chain over the real DEM of the lower Taquari
13	Table 13.1	Book	the same origin, measured and cached in `figuras/dados/limiar-drenagem.csv`
13	The data	Line	Copernicus GLO-30, the clip the flood study of the line downloaded; the layers the figure draws live in `figuras/dados/terreno-taquari.npz`
13	References	Published	Nobre et al. (2011) for HAND, Barnes et al. (2014) for Priority-Flood
Chapter 14 *The envelope: the agreement raster*			
14	The measurement	Application	`measure` in `sidecar/flood.py`
14	The raster	Application	`agreement_raster`, in the same file
14	The index	Application	the `iou` function, with the note on CSI
14	The ring	Application	`inset_mask` and `resolve_inset_margin`
14	The thresholds	Application	`THRESHOLDS_M` and `REFERENCE_THRESHOLD_M`
14	Drainage and ring	Application	`DRAINAGE_REF_KM2` and `INSET_MARGIN_M`
14	The qualifier	Application	`qualifier_text`, on the swap of SRTMGL1 for `alos-dem`
14	Figure 14.1	Book	`figuras/figuras.py`, which fetches the four products through `sidecar/dem.py` and runs `flood.measure`
14	Its cache	Book	`figuras/dados/envoltoria-taquari.npz`
14	The study of origin	Line	`E-hand-flood-baseline`, of the *TERRA-Research* line
Chapter 15 *Solar resource at a point*			
15	The action	Application	`action_solar_resource` in `sidecar/infer.py`
15	The chain	Application	`sidecar/solar.py`, whose opening declares the grid and the yield calibration
15	The series	Application	`fetch`, `annual_totals`, `monthly_climatology` and `clear_sky_index`
15	The geometry	Application	`prepare_hourly`, `transpose` and `sweep_tilt`
15	The yield	Application	`pv_yield_frame` and `modelled_performance_ratio`

Table B.2, continued

Ch.	What	Where it lives	Where it is
15	The cache	Application	`cached_power_series` and `power_cell_key`, in `sidecar/infer.py`
15	The study of origin	Line	the solar resource study of the *TERRA-Research* line, over three properties in western Paraná
15	Figures 15.1 and 15.2	Book	`figuras/figuras.py`, which reads the climatology and comparison tables of that study
15	Table 15.1	Book	`mede_tolerancia_inclinacao` in the same file, over the sweep of the study
15	References	Published	Perez et al. (1990) for the transposition, Faiman (2008) for the module temperature, Dobos (2014) for the photovoltaic model, Marion (2017) for diffuse incidence, Holmgren et al. (2018) for the implementations, Kopp and Lean (2011) for the solar constant, Doelling et al. (2016) for the radiation and Gelaro et al. (2017) for the meteorology

Chapter 16 *The energy model and the loss waterfall*

Ch.	What	Where it lives	Where it is
16	The action	Application	`action_energy_model` in `sidecar/infer.py`
16	The module	Application	`sidecar/energy.py`, which brings no new data source and consumes the chain of chapter 15
16	The three ratios	Application	`resolve_performance_ratio` and `modelled_factors`. The boundary of each product is guarded by `_require_resolved`
16	The waterfall	Application	`loss_waterfall`
16	The degradation	Application	`degradation_factor`
16	The tracking	Application	`tracking_comparison`, `_published_land_use` and `_parity_ground_coverage`
16	The profile	Application	`generation_profile`
16	The density	Application	`CAPACITY_DENSITY_BASES` and `resolve_capacity_density`
16	The exceedance	Application	`exceedance_table`
16	The area and the plant	Application	`plant_energy`, `largest_contiguous_area_ha` and `_applied_shading_derate`
16	Figures 16.1 and 16.2	Book	`figuras/mede_energia.py`, which runs the chain over the hourly series stored by the solar resource study of the *TERRA-Research* line, and `figuras/figuras.py`, which draws the result
16	References	Published	Dobos (2014) for the photovoltaic model and the loss terms, Faiman (2008) for the module temperature, Jordan and Kurtz (2013) for the degradation rate, Bolinger and Bolinger (2022) and Ong et al. (2013) for the capacity and energy densities per area, and the standard `IEC 61724-1` for the performance ratio convention, which defines it against the irradiance in the plane of the module